

Contour Clustering

Manual

Constantijn Kaland
Linguistics I – Phonetics and Phonology
University of Düsseldorf

This manual provides usage instructions for the R shiny application Contour Clustering (introduced in Kaland, 2021), which offers a graphical user interface to perform (cluster) analyses on f0 and other acoustic cues to speech. The application, scripts, tutorials, the latest version of this manual, and all other supplementary materials can be downloaded on the website:

<https://constantijnkaland.github.io/contourclustering/>



September 2026

Contents

1	Tips to get started	4
2	Requirements	5
2.1	Unit of analysis	5
2.2	Dataset	5
2.3	Segmentation	6
3	Workflow	8
3.1	Run app	8
4	Data	11
4.1	Load file	11
4.2	Read folder	12
4.3	Acoustics	13
4.3.1	F0: floor and ceiling	13
4.3.2	F0: fit	13
4.3.3	F0: smoothing bandwidth	14
4.3.4	F0: time-step	14
4.3.5	Number of measurement points	15
4.3.6	Intensity	16
4.3.7	Duration	16
4.3.8	Formants	17
4.3.9	Sampling	20
4.4	Speakers	21
4.5	Clean	21
4.6	Representations	23
4.6.1	F0	23
4.6.2	Intensity	24

4.6.3	Duration	25
4.6.4	Formants	25
5	Clustering	26
5.1	Variables for clustering	26
5.2	Distance metric	27
5.3	Clustering method	28
5.3.1	Hierarchical agglomerative clustering - HAC	28
5.3.2	Partitioning around medoids - PAM	31
5.3.3	Vowel centroids	32
5.4	Number of clusters	35
5.5	Plot settings	35
5.6	Cluster outputs	36
5.6.1	Dendrogram (HAC only)	36
5.6.2	Plot	37
5.6.3	Table	37
5.7	Subsetting	39
5.8	Evaluation	40
5.9	Prototypes	42
6	Data conventions	44
6.1	df.u	44
6.2	df.g	44
6.3	df.l	44
6.3.1	Singleton vs. time-series	45
6.3.2	Decimals	46
6.4	df.e	46
6.5	df.v	46
7	Settings	47
7.1	Audio conversion	47
7.2	File upload limit	47
7.3	Number of rows to display	48
7.4	Saving	48
7.5	Line aesthetics	48
7.6	Keeping objects	48
7.7	Garbage collection	49
7.8	Logfile	49

8	Software statements	50
8.1	Data protection	50
8.1.1	Citation	50
8.2	License	51
9	Appendix A: License	53
10	References	62
11	Index	67

1. Tips to get started

- download the latest version of the app
- use the index in this manual to jump to specific topics
- update R/-Studio and required packages (see versions in code)
- take a look at the workflow chart on page 10
- practice with a training dataset from the website
- hover your mouse on any location in the app to get tips
- don't forget to save your work in the app ('Save' or 'Save this' buttons)
- be inspired by previous work on various languages
- contribute by reporting errors, suggestions for new features, or your own work

2. Requirements

First, the three minimal requirements are discussed, which need to be fulfilled before contour clustering can be applied. A basic understanding of scripting R (R Core Team, 2025; R Studio Team, 2025) is helpful, but not required.

2.1 Unit of analysis

The researcher is required to decide which linguistic unit of analysis needs to be investigated. The unit of analysis depends on which level of prosody is subject to the interest of the researcher (e.g., word or phrase level). In principle, contour clustering can be applied to any interval extracted from speech. For example, for the investigation of lexical tone the unit of analysis would typically be the syllable or word, whereas for the investigation of phrase level intonation the unit of analysis would typically be the (intermediate) phrase. Note that some understanding of the prosodic relevance of these units in the language under investigation needs to be present. It is beyond the scope of this manual to provide fieldwork paradigms that help to identify the units that are relevant in the prosody of certain language (see e.g., Jun and Fletcher, 2014 for guidelines). Nespor and Vogel (2007) provide a phonological hierarchy that could be maintained. The decision for a unit has a couple of implications for later steps in the procedure. For this reason, it is advisable to keep the unit of analysis the same throughout one analysis, i.e. not comparing syllable-level f_0 movements with those found in phrases (Section 4.2).

2.2 Dataset

The researcher should have access to a collection of audio-recordings and corresponding annotations in Praat textgrids (Section 2.3 for annotation requirements), consisting of sufficient observations. Wave files should be in PCM-format (8, 16 or

24 bits per sample). Floating-point rates are not supported. The app offers a quick audio conversion option in the ‘Settings’ tab (Section 7) to obtain a readable audio format.

Praat textgrids can be exported from several programs, such as ELAN (Max Planck Institute for Psycholinguistics, 2022, see <https://www.mpi.nl/corpus/html/elan/ch01s04.html>) and STx (Noll et al., 2019, see <https://projects.ari.oeaw.ac.at/stx>).

What a sufficient amount of observations is, depends on the quality of the recordings and a number of later steps in the analysis. For example, when only four clusters need to be identified, the analysis might reach a representative outcome with less observations than when eight clusters need to be identified (see Section 5.4 for instructions on the number of clusters). It is possible that some data needs to be discarded due to measurement errors. It is therefore recommended to collect more units than strictly speaking needed. In the phrase contour example (Kaland, 2021, Section 3), 321 contours were used for initial analysis, on the basis of which at least one clearly defined follow-up hypothesis could be formulated concerning potential functional differences between in the f0 contours. For the tonal analysis (Kaland, 2021, Section 4) the contours of 213 syllables were analysed and again provided insightful results for further testing. Another indication of whether the dataset is large enough for cluster analysis can be obtained from the mean standard errors within each cluster (provided in the output table, Section 5.6). Large standard errors indicate large variance within the cluster, which could be the result of (too) few observations. It is important to note that in general, more observations will often lead to more representative results and that the ideal size of the dataset largely depends on decisions taken during the analysis (see Section 5.4 and 5.7). Unbalanced datasets are not necessarily problematic for the cluster analysis. For example, a dataset with 5% questions and 95% statements could still be accurately clustered when their differences in the acoustic cue(s) (i.e., f0) are clear enough.

2.3 Segmentation

Once the unit of analysis and the dataset are decided upon, the researcher needs to segment all units of analysis on an interval tier in a Praat textgrid (Boersma and Weenink, 2025). This means that the left and right boundary of each unit (an interval) need to be indicated. It is highly recommended to label the units using glosses in the language under investigation or another common language (e.g., English), or ideally both. For purely exploratory purposes, contour clustering does not require text labels for the intervals on the textgrid tier. The more information is available in

the textgrid for each unit, the more solid the interpretation of the results can be after analysis (see ‘contingency table’ in Section 5.6.3). This information can include other linguistic aspects that are not necessary included in the initial intonation analysis (e.g. segments, syllables, words, phrases, discourse units, grammatical annotations). See <https://osf.io/cr4vn/> for Praat scripts that handle and count (overlapping) intervals on multiple tiers.

To speed up segmentation, forced aligners can be used to obtain the boundaries of the relevant intervals. Pipelines such as Webmaus (Kisler, Reichel, and Schiel, 2017) or the Montreal Forced Aligner (MFA) (McAuliffe et al., 2017) are popular options. Both offer existing language models that provide high-quality alignments and sufficient options to configure the output. The alignments can be obtained as textgrids in both cases.

Webmaus: <https://clarin.phonetik.uni-muenchen.de/BASWebServices/interface/WebMAUSBasic>

Montreal Forced Aligner: <https://montreal-forced-aligner.readthedocs.io/en/latest/index.html>

3. Workflow

Figure 3.1 provides a schematic overview of the contour clustering workflow when using the application. The workflow is divided into a ‘data’ stage (Chapter 4) and ‘clustering’ stage (Chapter 5). The names of the respective tab panels in the application are used as section titles in the following, in order to provide a transparent reference to the location in the workflow as well as the in the application.

The data stage covers all operations that are required to prepare a dataset for clustering. Once that stage is completed, the clustering stage becomes available and concerns all operations related to the performance and processing of cluster analysis and its results.

In what follows, each step in the workflow is explained in a separate section, in the order of the workflow.

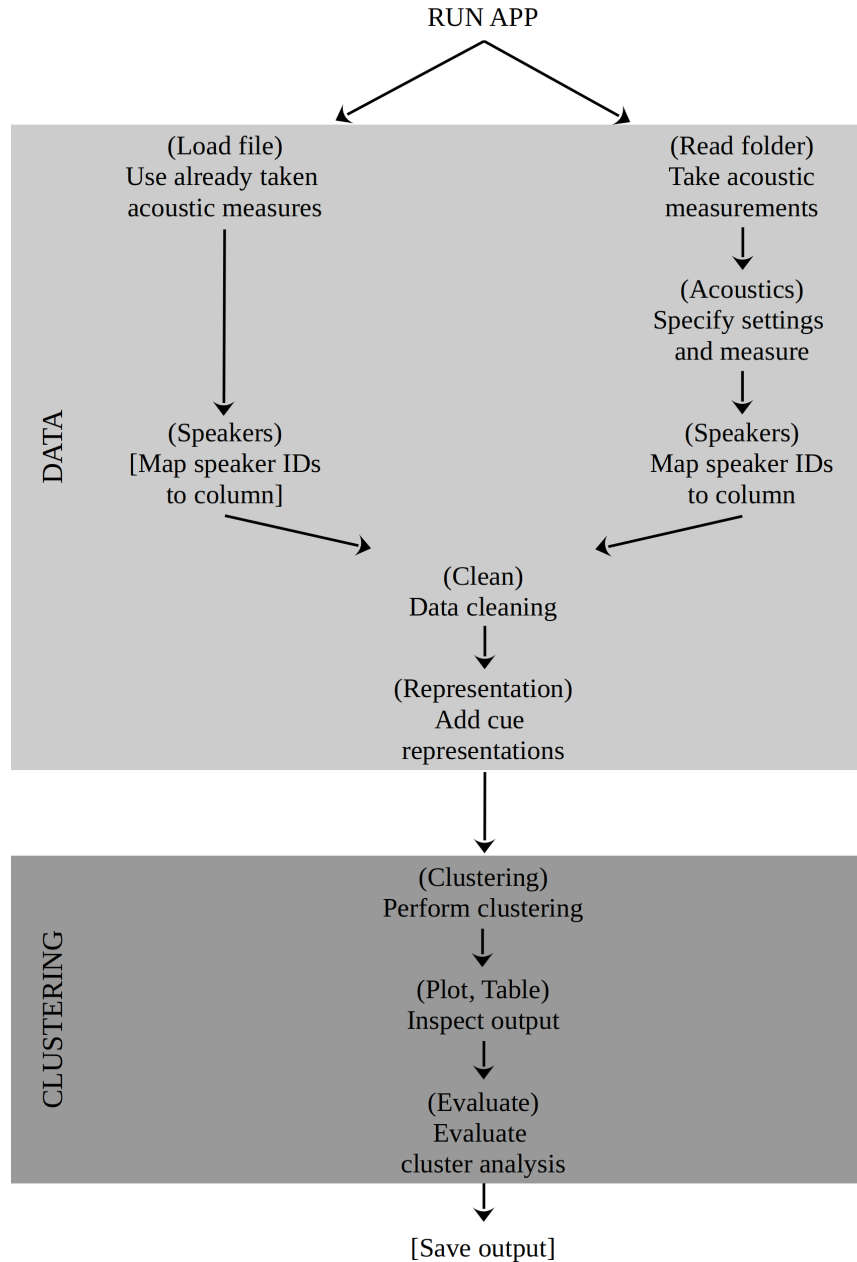
3.1 Run app

In order to run the application, R (R Core Team, 2025) and R Studio (R Studio Team, 2025) need to be installed. If not already installed, the application will install each required package automatically when the code is executed. If any packages needed a (new) installation, the R session will automatically restart and the application need to be started again. If no ‘Run app’ button is displayed, the `shiny` package (Chang et al., 2012) needs to be installed first (run: `install.packages("shiny")`). An overview of the required packages is provided in the comments section of the application code.

The application is best displayed in a maximized window. Any smaller app window might cause layout issues or controls being (partially) invisible. Windows users need to set the screen scaling manually to 100% in order to have access to all controls in the application (or use a command-line tool `SetDPI` available here: <https://github.com/immiko/SetDPI>). It is also possible to run the application in a browser window by choosing ‘Run external’ in R Studio or set the `launch.browser`

= T option in the code (commented out in the code by default).

Figure 3.1: Schematic overview of the Contour Clustering workflow, divided in a ‘Data’ and a ‘Clustering’ stage referring to the respective main tabpanels in the application. For each step, the names of sub-tabpanels are given between brackets. Square brackets indicate optional steps in the workflow.



4. Data

Once the app has started, a choice needs to be made about the source of the to be clustered data (requirements see Section 2.2): either upload an existing dataset (Load file - Section 4.1) or read audio files and textgrids from a folder (Read folder - Section 4.2).

4.1 Load file

Use this option if acoustic measures are already taken. Files that can be uploaded may be obtained from the application in a previous session, from running the ‘time-series.f0.praat’ script in Praat (Boersma and Weenink, 2025), or from (manually) compiled data by another source and/or the researcher. Note that non-locally stored files (server/cloud) may not be readable without copying them to the local hard disk.

Uploading a file requires a .csv file in ‘long’ format. This means that each row in the data represents a single measurement point (i.e., a single f0 value in Hertz). Each row needs to have values in at least the following columns:

Table 4.1: Required columns when uploading a data file.

Column name	Class	Content
filename	character	Refers to the filename (.wav/.mp3) from which the measurements were taken.
start	numeric	Start time of the interval (seconds)
end	numeric	End time of the interval (seconds)
speaker	character	Speaker IDs (can be identical to or extracted from another character column)
interval_label	character	Text referring to the interval (usually taken from textgrid intervals)
stepnumber	integer	The sequence number of the measurement point in the time-series
f0	numeric	Fundamental frequency measured in Hertz, ERB or semitone (app can convert)

Additional columns can be taken from the uploaded datafile. The application caters for columns that were used in older versions (e.g., ‘jumpkilleffect’, ‘steptime’),

as well as ones that represent other acoustic cues ('intensity' or 'duration') or additional columns that require manual specification of whether they contain character values, numeric values, or are referring to an (acoustic) variable that should be taken into account for clustering ('cc', see Section 6).

It is furthermore possible to subset the uploaded data for specific filenames and/or interval labels.

4.2 Read folder

The 'Read folder' tab panel should be selected if no acoustic measurements were taken yet. The path to a folder that contains both audio files and textgrids needs to be given, and the extension of the audio files (.wav or .mp3) specified. Note that non-locally stored files (server/cloud) may not be readable.

The first audio file in the folder will be checked for its quality. If the sampling rate is above 32 kHz, the bit depth higher than 16 bits or the channels are stereo, a suggestion is given to convert the audio files. This is necessary when file-reading errors are encountered and recommended to speed up processing (Section 7.1).

Reading the folder means that the application scans for audio files and textgrids that have identical names (only differ in extension, e.g., 'Recording_1.wav' and 'Recording_1.TextGrid'). The resulting list of names is returned and individual files can be (de)selected. The audio files are processed faster when they have only one channel (mono). The textgrid files should have the file encoding UTF-8 or ASCII and should not be saved as 'short text file' in Praat. Textgrids not conforming to those conditions cannot be read by the `readtextgrid` package (Mahr, 2020) and cause the application to halt. A warning message is given that shows which file was processed when the error occurred, in order to facilitate debugging the textgrid(s). Note that the application does not automatically convert the file encoding as to avoid losing the use of specific (special) characters in the textgrids. It is up to the user to provide (a converted copy of) the textgrids in a readable encoding.

In addition, textgrids need to have boundaries at the start and end of each interval. This may be trivial, but in Praat (Boersma and Weenink, 2025) it is possible to have intervals at the start and end of tiers for which the left or right boundary respectively are not set explicitly (i.e., coincide with the audio file start or end). These non-explicit boundaries will cause measurement problems in the application and are therefore automatically corrected if they are detected by shifting them 1 ms forward (at the start) or backward (at the end) so that they fall *within* the audio file duration and not coincide with the edges. A warning notification is given when

selected tier(s) have coinciding boundaries. To avoid automatic correction of interval boundaries, textgrids need to be revised such that all intervals on the selected tier have a start *and* end boundary.

All interval tiers found in the textgrids can be selected for further processing. In case multiple tiers are selected, they should be of comparable units of analysis. That is, it is not advised to feed a tier with segmented syllables and a tier with segmented phrases into the same analysis. With multiple tiers selected, the name of each tier is prepended to the text in the ‘interval_label’ column. The data read from the textgrids (`readtextgrid::read_textgrid()`, Mahr, 2020) can be viewed in the ‘Data (TextGrids)’ tab.

It is furthermore possible to select specific interval labels from the selected tier(s) for further processing. It is recommended to make selections as early as reading the folder, in order to avoid processing of unwanted data, which may significantly reduce processing time.

4.3 Acoustics

The ‘Acoustics’ tab provides an environment to take acoustic measurements of f0, intensity, duration and formants - only available after ‘Read folder’ (Figure 3.1). Each cue can be (de)selected depending whether it should become available for cluster analysis. Their respective settings are discussed in the following.

4.3.1 F0: floor and ceiling

The f0 tracking method is Modified Harmonic Sieve (Scheffers, 1983 as implemented in `wrassp::mhsF0()`, Winkelmann et al., 2023). This method requires an f0 window for the expected values, as defined by the f0 floor and f0 ceiling. These can be adjusted depending on the voice under analysis or the type of phonation (Gordon and Ladefoged, 2001).

4.3.2 F0: fit

The Modified Harmonic Sieve method aims at detecting f0 in noisy environments by a maximum likelihood estimation from its harmonics (Goldstein, 1973). The f0 fit threshold specifies the minimum probability for an f0 measurement to fit. Increasing this value makes the detection algorithm more strict, rejecting more f0 candidates and accepting only the better fitting ones (more accuracy, less candidates). Decreasing this value will accept more candidates at the expense of tracking accuracy.

4.3.3 F0: smoothing bandwidth

By default, the f0 measures are interpolated and extrapolated, which means that all missing f0 points are filled in by linear interpolation and missing f0 points at the edges are filled in by extrapolation using a constant based on the first/last available f0 point. This is done to obtain uninterrupted f0 contours, which is at least partially motivated by perception research (Mixdorff and Niebuhr, 2013).

The f0 contour is then smoothed using kernel density estimation (KDE). The applied method is described in Silverman (1986). The smoothing bandwidth positively correlates with the degree of smoothing (i.e., larger bandwidth = more smoothing). The bandwidth value is the input of the `stats::ksmooth()` function. The smoothing allows to reduce the influence of local (erroneous or irrelevant) f0 perturbations on the overall contour. It is recommended to try out multiple settings and inspect the results using the sampling method (Section 4.3.9).

4.3.4 F0: time-step

The time-step setting refers to the frame duration used to calculate f0 (`wrassp::mhsF0()` ‘windowShift = ’). The standard setting used in Praat depends on the pitch minimum (or pitch floor; see Praat Manual “Sound: To Pitch...”; Boersma and Weenink, 2025) and defaults to 10 ms with a minimum of 75 Hz. Manually specifying the time-step provides a way to configure the resolution of the f0 measures. This might be needed in addition to number of measures and smoothing. Note that the minimum interval duration required to measure f0 is set to twice the time-step (default: 20 ms). This is done to obtain a minimum amount of speech material for f0 tracking. Lowering the time-step value may thus allow to take f0 measures from smaller intervals.

The time-step setting determines how many measurement points are initially taken for a contour, which at the stage of f0 tracking is different from the number-of-measurement-points setting (Section 4.3.5). For example, for a 200 ms interval one obtains 20 f0 measures with a 10 ms time-step and one obtains 40 f0 measures with a 5 ms time-step. This number of measurement points will be converted to the number of measurement points the user sets to estimate each contour (Section 4.3.5), i.e., after inter-, extra-polation and smoothing. If the final number of measurement points is more than the number of points obtained using the time-step setting, the likelihood of inaccurate f0 approximation increases. That is, a contour obtained from 20 measures (e.g. 200 ms interval, 10 ms time-step) will not become more accurate when setting the number of measurement points to 40. Approaching this issue the other way around is recommended, i.e., setting the time-step such that the number

of f0 tracking points is larger than the chosen number of measurement points. Thus, the time-step setting offers a way to safeguard that the tracking resolution equals or exceeds the contour resolution. Some additional lines of text, dependent on the data and the settings, are provided in the application to guide this process.

4.3.5 Number of measurement points

The researcher needs to decide how many f0 measurements are needed to represent the smoothed contour as time-series. It is important to realise that the smoothed contour is already the product of interpolation, extrapolation and smoothing, potentially based on just a few tracked f0 points. Representing the smoothed contour by more points than actually tracked, could be misleading as this potentially generates the illusion of accurate representation (e.g., representing a smoothed f0 contour by 20 points when it is based on 5 tracked f0 points, see Section 4.3.4).

It is important that eventual turning points in the contour are well represented by both the smoothing and the number of measurement points. This depends largely on the unit of analysis and the desired accuracy of the measures. Taking too few measures could lead to missing essential f0 movements within the contour, whereas too many measures could give too much importance to insignificantly small f0 measures, although the latter should be largely taken care of by the smoothing process. Note that the smoothing process (Section 4.3.3) could also be responsible for the over-estimation of small f0 movements. It is recommended to change the number of measurement points only in accordance with the smoothing resolution, whilst considering the output of the cluster analysis (Section 5.6).

It is important to note that inaccurate f0 measurements are likely to occur in each dataset. The cluster analysis provides an automatic detection of erroneous and outlying contours, which can then be removed from the data (Section 5.7). Changing the number of measures to account for a small number of erroneous contours is not recommended, as this affects also the correct measures. Only if the same type of f0 errors occur repeatedly, changing the number of measures could be considered. The sampling stage (Section 4.3.9) provide an important way to overcome the most severe tracking problems. An example of the sampling window for f0 measurements is given in Figure 4.1.

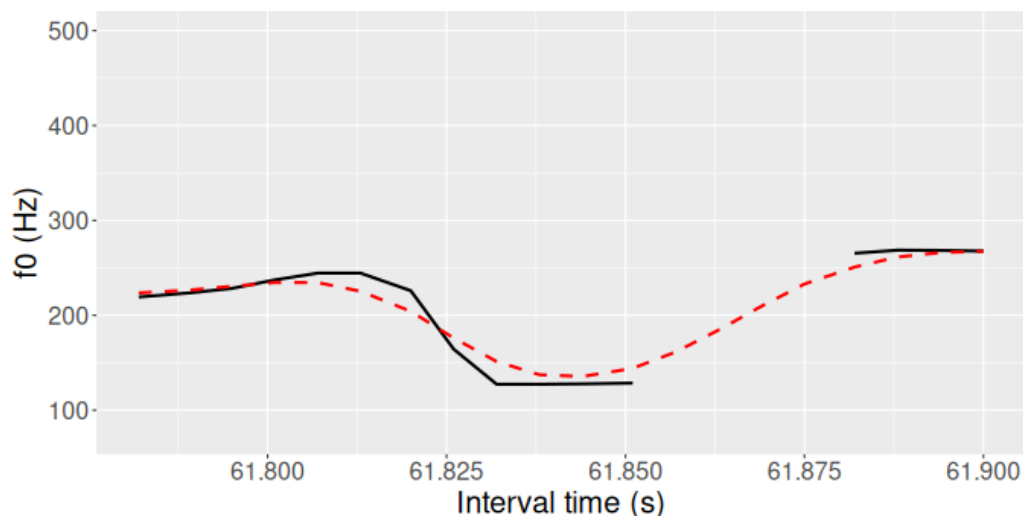


Figure 4.1: Example of an f_0 contour with the tracked values in black and the interpolated, extrapolated and smoothed values in red (dashed). For this example, the f_0 fit was set to 0.6, the smoothing bandwidth to 4, time-step to 10 ms, and the number of measurement points to 20.

4.3.6 Intensity

Intensity (short-term Root Mean Square amplitude in dB) can be measured as additional (or sole) acoustic cue. The intensity measures are taken using the `wrassp:rmsana()` function (Winkelmann et al., 2023). The number of measurement points is identical for intensity and f_0 time-series.

4.3.7 Duration

Duration measurements can be added for clustering and will be taken as singleton ones (not time-series) and provide a way to represent the temporal information that is lost when representing each time-series with the same number of measurement points (time-warping). The duration is calculated by subtracting the interval end time from the interval start time in seconds. This is done once per interval (hence ‘singleton’, see Section 6.3.1) and written to the first row in the data that represents this interval (i.e., for stepnumber 1).

4.3.8 Formants

Formants can be estimated and smoothed as time-series using the Split-Levinson Algorithm (SLA) (Willems, 1987) as implemented in the `wrassp::forest()` function (Winkelmann et al., 2023). The number of formants (maximum 5) and the number of measurement points can be determined by the user. The smoothing bandwidth determines how closely the tracked (and inter- and extrapolated formant contour is followed in the time-series that ultimately represents the formant for clustering. This is done in the same way as described for f_0 (Section 4.3.3).

The boundaries specified in the `textgrid` (saved as ‘df.g’, Section 6) are taken as the boundaries of the vowel interval. Depending on how these boundaries are set, they might specify a too wide interval, or might consist of voiceless material. To improve the accuracy of the formant tracking method, the application offers a way to track formants (using the SLA) in a specific portion of the interval (henceforth: subinterval). The determination of this subinterval is based on intensity peaks and also described in detail in Kaland (2019, p.61/62). The approach is based on the idea that formants are most stable and most reliably measured for voiced material with sufficient intensity. The validity of this assumption can be easily observed by simultaneous visual inspection of the spectrogram and intensity contour (i.e., in the Praat editor). This approach also mitigates that the SLA always returns formant values, regardless of how salient the formants actually are (as mentioned in https://fon.hum.uva.nl/praat/manual/Sound_To_Formant_sl_.html). An example from the application’s sampling window (Section 4.3.9) is given in Figure 4.2.

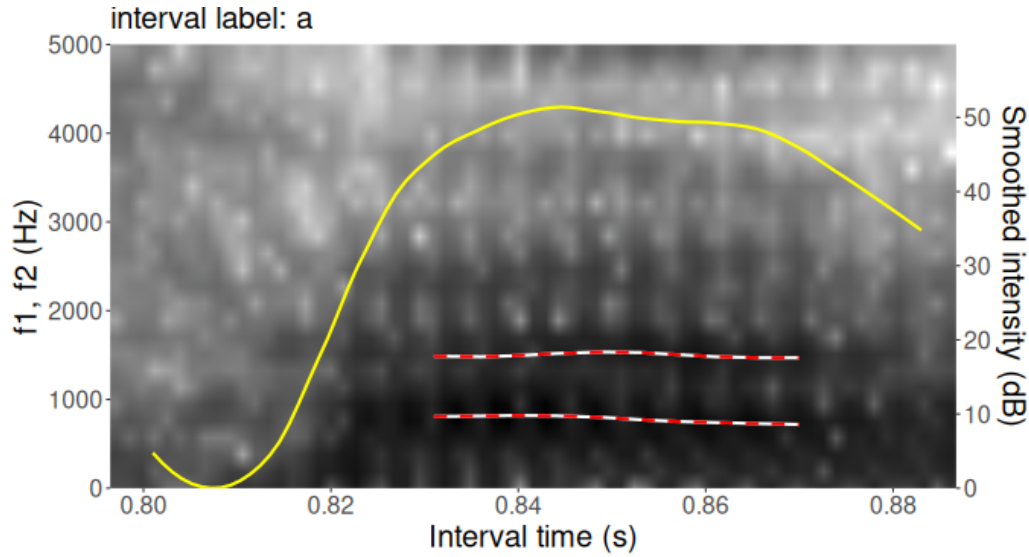


Figure 4.2: Example of a dynamic sub-interval used to track the first two formants of [a]. The white line represents the tracked formant values, the red dashed line represents the smoothed formant values (which overlap with tracked formant values in this example), the yellow line represents the smoothed intensity levels. The dynamic subinterval for formant estimation was obtained by allowing a 4% drop on either side of the intensity peak.

In order to find the optimal subinterval for formant estimation, the application first determines where the intensity peak occurs in the interval as specified in the textgrid. After the timestamp of the intensity peak is determined, a search is initiated for the timestamp at which the intensity level first drops below a set threshold, in either direction from the peak. This threshold is defined as a drop in intensity relative to the peak (e.g., 4% in Kaland, 2019 and 5% in Culhane and Kaland, 2025) and can be set by the user. The timestamps corresponding to the allowed relative intensity drop on either side of the peak are taken as the edges of the subinterval for which formant estimation is carried out. The intensity levels are measured using the method described in Section 4.3.6. A single peak is used for the search for the subinterval edges. In case multiple intensity peaks occur, only the peak with the highest intensity level is used for the search. A highly wiggly intensity curve could be a sign of the interval being too large or too small for formant estimation, and/or too much noise hinders the detection of a reliable subinterval for formant tracking.

Note that the use of a relative threshold (i.e., a percentage) will find the optimal

subinterval for formant estimation in a dynamic way. That is, the approach does not depend on the exact dB level of the recording, and for intensity curves with a plateau, the subinterval for formant estimation is expectedly larger than for intensity contours that peak more steeply. If no value below the specified threshold is found within the interval, the outer edge of the interval will correspond to the edge of the formant tracking subinterval. When using a dynamic subinterval around the intensity peak, the time-series formant values are displayed in the plots of the sampling window at the relevant (sub-)intervals. Note that this time-series (sub-)interval will be approximated using the set number of measurement points, regardless of its original length. It is therefore recommended to adjust the smoothing bandwidth accordingly, if the aim is to avoid wiggly formants.

The intensity-peak method is particularly useful when the interval (i.e., unit of analysis) might consist of sounds in which no formants are expected, such as the entire syllable including consonants or a complete vowel interval including undesired co-articulation effects. If formant transitions are to be tracked, e.g., to analyse co-articulation effects or diphthongs, it is advised to take tighter intervals (i.e., just vowel portions) and relax the relative threshold by setting it to a higher value. In this way the dynamic subinterval is larger on average and better able to capture entire formant transitions.

Tracking accuracy is key in formant estimation. When dealing with speech data for which the sampling keeps estimating formants in an unreliable way, it is advised to choose a singleton value as formant representation. The application offers: formant values at the intensity peak, the mean of the time-series, or the median of the time-series. Note that the singleton formant values taken at the intensity peak do not rely on time-series estimation, unlike the mean and median. The latter two might be chosen when time-series show too much wiggleness that cannot be overcome by changing the smoothing bandwidth. Both time-series and singleton formant representations can be used for clustering. Plotting is restricted to time-series values (see Section 5.6).

To further improve the accuracy formant estimation, the length of the vocal tract of the speaker can be specified (`gender` option in the `wrassp::forest()` function; Winkelman et al., 2023). Typically, female speakers have shorter vocal tracts than male speakers. The formant tracking function offers three options: short vocal tracts, long vocal tracts and 'unknown' (the default). If the vocal tract length is known, this can be specified for all at once, or based on the text string found in the filename, tier name or interval label. Text strings can be split by any punctuation mark found in that string and any of the respective sub-strings can be selected as containing the vocal tract length information. Each unique sub-string found can be selected as vocal

tract length marker for each length option separately, even if multiple apply. The mapping of vocal tract length information adds a column `vc.length` to `df.g` with the labels `s`, `l` or `u` for short, long and unknown respectively. This information can later be added to the respective speaker IDs (Section 4.4). Note that a systematic and consistent naming of files, tiers or intervals is important for mapping vocal tract length information in the application.

4.3.9 Sampling

Before taking f0 or formant measures of an entire dataset, there is an option to sample a specified number of randomly chosen intervals to inspect the accuracy of the acoustic settings. In particular the number of measurement points and smoothing bandwidth should be changed to improve the quality of the measurements, in connection to the specific settings for the chosen acoustic measure (f0 fit, formant dynamic window). It is likely that the standard settings produce inaccurately tracked contours and the settings should be adjusted accordingly. After each sampling, plots are generated showing the tracked points using the tracking method (black solid line) and the interpolated, extrapolated and smoothed f0 smoothing contour (red dashed line). Sampling provides therefore an essential step before taking acoustic measures for all observations in the data. It is possible that the acoustic measure cannot be tracked (accurately) in all intervals; this will be shown in the sampling plots and NAs will be written to the output file. In addition to the plot, the audio taken from the sample can be played. The sampled interval(s) are cut from the original file and stored temporarily in the ‘www’ folder in the folder from which the application is run. The ‘www’ folder and its contents will be automatically removed when sampling has finished.

Each time ‘Visualize sample’ is clicked, the current settings are used to generate the plots based on the sample that was already taken. Only when ‘Take sample’ is clicked, a new sample is taken. Changing the size of the sample is only effectuated when ‘Take sample’ is clicked.

4.4 Speakers

When taking acoustic measurements in the application or when the uploaded data-file does not have a column with speaker IDs, the tab ‘Speakers’ offers a way to create a speaker column. This can be a 1:1 copy of another character column that is already present in the data or an extracted string from the any character column in the data. The latter option is useful when, for example, the ‘interval_label’ column consists of multiple sources of information, e.g., task number, speaker ID and condition in the format `008_pp3_test`. The application offers a way to split the string according to any punctuation mark found in that column (regex class: `[:punct:]`). In the example, an underscore would be the separator to extract ‘pp3’ as speaker ID.

The extraction is done in three stages, from left to right in the ‘Speakers’ tab:

- 1) select the column containing the speaker IDs
- 2) select the separator character
- 3) select the column with the speaker IDs after separation.

The selection made in step 3 can be taken as the new column ‘speaker’. Note that the number of unique character strings found in the (new) ‘Speaker’ column equals the number of speakers. If vocal tract length information was specified (Section 4.3.8), the respective labels can optionally be appended to the speaker IDs.

4.5 Clean

The cleaning tab is presented once the dataset has all required columns and at least one column with acoustic measures for clustering (commonly f0 values). Upon display, the ‘Clean’ tab shows a summary of the number of filenames, speakers, measurement points and intervals (contours), and any problems currently detected in the data. Table 4.2 lists the most common problems that are checked.

Table 4.2: Problems checked by the cleaning process

Problem	Description
missing filenames	NA entries in ‘filename’ column
missing speaker IDs	NA entries in ‘speaker’ column
empty interval labels	accidental spaces ‘ ’ in ‘interval_label’ column
missing f0	NA entries in ‘f0’ column
negative f0	negative value in ‘f0’ column (e.g., result of previous standardization)
rate of f0 change	checks adjacent f0 values against rates in Xu and Sun (2002, Table X)

Any detected problems will be listed in the summary and can be solved by clicking ‘Clean and re-check’, after which a new summary is given. It is important to observe how many observations get removed in the cleaning process, to see whether there are any structural problems in the data.

With regard to the maximum rate of f0 change (velocity), the option is provided to allow a maximum number of adjacent f0 values (measurement points) in a contour to exceed the rates in Xu and Sun (2002, Table X): 72 ST/s for rises and 96 ST/s for falls. Note that exceedings might occur as a result of (insufficient) smoothing. Allowing some exceedings in a contour could be considered as a means to preserve data that would otherwise be discarded for clustering. For inspection, the error summary reports the number of contours times the number of exceedings. In order to minimize the number of exceedings, it is recommended to obtain more reliable measurements, e.g., by increasing the number of measurement points with an increased f0 fit and/or a higher smoothing bandwidth.

For manual inspection of the contours with exceedings, use the ‘Save’ option in the ‘Data’ tab. Saving ‘df.l’ immediately after the error summary (N.B. before cleaning), will save the data with an additional column ‘d.f0’, which gives the number of exceedings for each contour. Exceedings are marked with ‘1’ from the second measurement point onward (stepnumber=2), because the exceedings are calculated between the measurement point and its previous neighbour. The total number of exceedings for a contour are given at stepnumber=1.

Although it is possible to ignore the detected problems and continue without cleaning, this option is not recommended and only advised if the application has detected problems that can be ignored. Ignoring detected problems may lead to application crashes.

4.6 Representations

The ‘Representation’ tab offers a final step of bringing the data in the desired shape for clustering. Arriving at this tab means that all source data is present and cleaned. It is therefore possible to continue directly to clustering. However, it might be desired to apply conversions or (speaker) corrections to the (raw) acoustic measurements in order to obtain additional and more representative values for the acoustic cues. This can be done for f0, intensity and duration separately, when present in the data. The options for the respective cues are discussed below. Clicking ‘Add .. representation’ adds the chosen representation as new values in an additional column to the dataset (see Section 6).

4.6.1 F0

All operations to obtain f0 representations can be combined, i.e. it is possible to do a functional principal component analysis on the acceleration (d2) values of speaker-standardized semitone values.

F0 scale can be set to Hertz, ERB or semitone. Semitones are converted with 50 Hertz as reference value. ERB conversion according to the formula in Greenwood (1961, p.1352, Eq. 2).

F0 values can be speaker corrected by means of standardization (z-scoring) or octave-median rescaling (De Looze and Hirst, 2014).

Derivatives (d1 - velocity, d2 - acceleration, d3 - jerk) can be taken from the (converted) (speaker-corrected) f0 values. See Kaland (2025) for a study on their informativeness.

Functional principal component analysis (fPCA, see Gubian, Torreira, and Boves, 2015) can be run on the (converted) (speaker-corrected) (derived) f0 values. The analysis is automatically run as soon as the number of principal components (PCs) is not zero. The output is a graph showing the harmonic values for each PC and settings are given for smoothing (λ - lambda) and which PC should be selected for clustering. It is possible to run fPCA with four components whilst adding only PC1 and PC2 to the data for clustering. The values of each PC appear in a new column as singleton measure, i.e., one value per interval at measurement point 1 (Section 6.3.1).

Standardizing the f0 subtracts the mean f0 a speaker from each f0 measure of that speaker and divides the outcome by the f0 standard deviation of that speaker. After standardizing, the mean f0 centers around zero and the standard deviation is one. The minimum and maximum f0 values still reflect the original distribution

and outlying f0 values are therefore not accounted for. Note that if outliers are the result of f0 measurement errors, they can still effectively be handled by means of subsetting (Section 5.7). Standardizing is a recommended method to preserve functional f0 differences such as tone contrasts when comparing different speakers (e.g. Rose, 1987).

Another method to scale the f0 range is based on the octave scale, taking into account the speaker’s median f0. This makes the f0 measures more robust to outliers and is proposed as more representative method for speech melody compared to reference values based on the mean and expressed in semitones (De Looze and Hirst, 2014). Octave-median scaling appeared particularly accurate for estimating acoustic emphasis as a result of focus, topic change or turn-taking.

fPCA reduces the dimensionality of the f0 contours to a chosen number of principal components. The components represent deformation functions relative to the overall mean f0 contour and can be visualized using curves (contours). Each principal component comes with scores for each contour, which allow for a reconstruction of the original f0 contour from the mean f0 contour + the principal component curves + scores. The application offers a basic fPCA functionality: setting the number of components, a plot of the component curves, and an overview of the proportion of the variance explained by each component. The user can select which component(s) need to be included into the cluster analysis. fPCA is performed using the `fda` package (Ramsay, 2003) following the procedure outlined in <https://github.com/caojiguo/FDAcourse2019>. Additional materials can be found on <https://github.com/uasolo/FPCA-phonetics-workshop>.

The application of cluster analysis on the fPCA scores offers a promising way of handling the f0 variation, possibly less affected by noise in the data. Note that the fPCA is run after any of the selected cleaning and/or speaker correction methods are applied. The user needs to select the number of principal components (default: 4) and λ (default: 1), the latter being a smoothing factor used in the function `fdPar()` to define a functional parameter object. Higher lambda numbers indicate more smoothing, which could lead to larger deviations from the original f0 contours in the reconstructed fPCA curves.

4.6.2 Intensity

Intensity values can be standardized (z-scored) per filename or per speaker. Filename standardization is useful when recordings (files) had different microphone levels. Speaker standardization standardizes over the speaker IDs and is useful when speakers’ voices differ in loudness.

4.6.3 Duration

Duration values can be standardized per speaker to account for individual differences in speech tempo (in the unit of analysis).

4.6.4 Formants

Formant values can be converted to Bark to obtain a perceptual realistic scale (formula from Traunmüller, 1990). Formant values can also be standardized per speaker, which equals the Lobanov-normalisation (Lobanov, 1971). Any conversion/standardization is applied to all formants one-by-one, regardless of the values being singleton or part of a time-series. Bark-converted formant values are required for vowel centroid clustering (Section 5.3).

5. Clustering

The clustering stage starts when all required data is present, has been cleaned and the desired representations for the acoustic cue(s) have been added to the data. The ‘Clustering’ tab provides cluster settings (side panel) and cluster outputs such as plots, tables and evaluations (main panel).

5.1 Variables for clustering

From the acoustic representations present in the data, any number can be chosen as variable(s) to apply cluster analysis to. When more than one variable is chosen, the cluster analysis will be *multivariate clustering*. This is handled by the application in the following way. First, a distance matrix is computed for each variable (with the chosen distance metric, see Section 5.2). The distance values in each distance matrix are then scaled such that they lie between zero and one. The scaled distances are summed into a single distance matrix, which is then taken as the input for cluster analysis.

Multivariate time-series clustering can in theory also be done by concatenating the time-series. Note, however, that this does not allow that combinations of time-series values (e.g., f0) and singleton (e.g., duration) values are given equal weight. In addition, concatenation is not suitable when using, e.g., dynamic time warping as distance metric, because all measures of all variables are analysed as a single vector per observation. The application thus uses the scaling-summing approach for multivariate clustering, giving equal weight to the distances of the respective variables.

See Section 5.2 and Section 6.3.1 for restrictions on the use of distance metrics when choosing the variable(s) for clustering.

5.2 Distance metric

The available distance metrics are Euclidean (L2 Norm / RMSE), mean absolute scaled error (MASE), dynamic time warping and Pearson correlation (see Table 5.1). See Kaland (2023) for a study on how well perceived contour similarity may be approximated by distance metrics.

Table 5.1: Overview of available distance metrics and the respective R package used for computation. Only metrics marked with * are suitable for singleton measures.

Distance metric	R-package	pros	cons
*Euclidean (L2 Norm = RMSE)	stats	tested on f0 contours	insensitive to misalignment
mean absolute scaled error (MASE)	Metrics	outperforms RMSD	not widely used for f0
*dynamic time warping	proxy	sensitive to misalignment	overcorrection
pearson correlation	proxy	overall similarity	contour cannot be flat

Euclidean distance (L2 Norm) or root mean squared error (RMSE) is widely used to compute the distance between two vectors due to its low computational cost and frequent availability. It is however, insensitive to misalignments in time as it is a lock-step measure and does not handle outliers well (Esling and Agon, 2012). This means that clustering outputs using Euclidean distance as a distance metric become more reliable when the data is cleaned and/or converted beforehand. This metric has been tested on f0 contours (Hermes, 1998) reflecting intonation perception to some extent.

Mean absolute squared error (MASE) is a time-series metric used to calculate the accuracy of weather forecasts (Hyndman and Koehler, 2006). This metric is scale-invariant and penalizes negative and positive distances in equal ways, outperforming related metrics such as root mean squared deviation (RMSD). Although the (f0) contour comparison is different from quantifying prediction accuracy - and currently needs further testing, the method is a potential alternative and applicable to time-series (not suitable for singleton measures). For each distance calculation between two contours, one contour is taken as if it were the actual weather observation and the other contour is taken as the prediction. The output MASE value can be taken as a distance metric as it shared the assumption that at zero, there is no distance between the contours (no errors between observation and prediction). MASE values greater than one indicate that a naive forecast (assuming no change) would preform better than the given forecast values. It is likely that f0 contour distances expressed in MASE end up being higher than one, indicating that one contour could not be

predicted from the other (i.e. are highly dissimilar). In this way, the MASE threshold of 1 could be used as cut-off point to potentially improve cluster accuracy.

Dynamic time-warping provides a way to account for misalignments between two contours that are otherwise similar in shape. These misalignments could have been the results of the time-warping that is done by taking a fixed number of measurement points to represent the f0 contour. Dynamic time warping as a distance metric should be avoided if the length of the unit of analysis has already been controlled for. That is, there is a trade-off to the usefulness of dynamic time-warping if all intervals have (almost) the same length. In this case, time-warping might analyse misaligned portions of the f0 contours as been similar although they are not. The importance of time-warping thus depends on the research question and data at hand.

Pearson correlation distance is an effective lock-step measure to compute overall similarity between contours that are represented by time-series data (not suitable for singleton measures). It is important that Pearson correlation coefficients are computed in such a way that they range from -1 to 1 (maximum negative and maximum positive correlation respectively). If direction of the f0 contour is not taken into account (as by taking the absolute correlation coefficient), the distance metrics will regard simultaneously diverging contours as being similar. The computation as implemented in the application takes direction into account and is therefore suitable for application to acoustic variables.

5.3 Clustering method

Two common clustering methods are available in the application: hierarchical agglomerative clustering (HAC) and partitioning around medoids (PAM). HAC and PAM differ fundamentally in their mathematical approach (further reading: Kaufman and Rousseeuw, 1990; Reynolds et al., 2006; Scitovski et al., 2021). A third method called vowel centroid clustering is specifically developed for the analysis of vowels in the application. Vowel centroid clustering is essentially a centroid based method such as PAM or K-means, with the crucial difference that centroids are not stabilized in the cluster process, but beforehand based on peaks in a two-dimensional density matrix (Section 5.3.3).

5.3.1 Hierarchical agglomerative clustering - HAC

For HAC there are multiple ways in which the clusters can be formed. Note that HAC in the current approach is bottom up, thus starting with each observation in a separate cluster. Clusters are formed by merging existing clusters until there are

only two clusters left. Theoretically, one extra step applies in that the final two clusters are merged such that all observations are in a single cluster. However, this final step is tantamount to no clustering and is not informative given that clustering aims at finding groups in a dataset. The dendrogram provides useful directions for the degree and type of subsetting needed, as further discussed in Section 5.6.1 and 5.7.

Which clusters get merged is determined by the *linkage criterion*. The linkage criterion specifies how distances (dissimilarities) between clusters are computed. An overview of the linkage criteria available in the application is given in Table 5.2. The default one recommended here is complete linkage, given the idea that maximal acoustic contrasts often underlie linguistically meaningful differences (e.g., the composition of vowel inventories, Lindblom, 1986). There is no a priori objection to select a different linkage criterion, depending on the type of f0 variation in the data.

Table 5.2: Overview of linkage criteria

Linkage criterion	Cluster dissimilarity definition
Complete	Computes all pairwise dissimilarities between the observation(s) in the clusters and takes the maximal dissimilarity. Clusters with the smallest maximal dissimilarity get merged, leading to maximal inter-cluster dissimilarity.
Single	Computes all pairwise dissimilarities between the observation(s) in the clusters and takes the minimal dissimilarity. Clusters with the smallest minimal dissimilarity get merged, leading to minimal inter-cluster dissimilarity.
Average (UPGMA)	Computes all pairwise dissimilarities between the observation(s) in the clusters and takes the mean dissimilarity. Clusters with the smallest mean dissimilarity get merged.
Ward	Computes the increase in sum of squares for all possible pairwise merges of clusters. Clusters with the smallest increase get merged. This method keeps the total within-cluster variance minimal. Ward.D and Ward.D2 differ in that the cluster differences are squared in the latter and therefore emphasized, leading to easier cluster differentiation.
McQuitty (WPGMA)	Similar method to average linkage, without considering the number of observations in a cluster and with taking into account the similarity between the most recently merged clusters.
Centroid (UPGMC)*	Computes the centroids (central point of all observations, i.e. a vector of means) of the clusters. Clusters with the centroids closest to each other get merged.
Median (WPGMC)*	Similar to centroid linkage, with taking into account the similarity between the most recently merged clusters.

* These linkage criteria have the risk of leading to inversion. For most link-

age criteria, merging happens on the basis of iteratively larger dissimilarities. With centroid-based linkage, centroid distances could get smaller in subsequent merging iterations. In such case, it is no longer possible to assume that with an increasing number of clusters variance among the observations in a cluster increases, whereas variance among the clusters decreases (see within and between cluster variance discussed in Section 5.8).

5.3.2 Partitioning around medoids - PAM

PAM (Kaufmann and Rousseeuw, 1987) makes use of medoids, which are actual observations that are taken as cluster ‘centers’ (BUILD stage). This is initially done by selecting k observations that have the smallest sum of distances to all other observations (with k being the number of clusters to be identified). Each (non-medoid) observation is then associated to the (according to the distance metric) nearest medoid. Then, PAM attempts to improve the clustering quality by reassigning medoids (SWAP stage). For each reassignment it is checked whether the average dissimilarity between the medoid and their associates decreases. If so, the reassignment is accepted. When the medoids do not change anymore, the partitioning of the data has arrived at a stable outcome, which is taken as the final cluster assignment.

PAM has been shown to be less sensitive for outlying contours than HAC (Kaland, 2025; Marchini, Steffman, and Kaland, 2025), which helps to obtain homogeneous clusters, but might ignore potentially relevant variation in the data.

To reduce computation cost, several PAM improvements have been proposed. The optimizers are therefore recommended for large datasets to optimize the processing speed. None of the optimizers changes the cluster assignment (unlike the linkage criterion in HAC). Table 5.3 lists the available optimizers in the application.

Table 5.3: Overview of PAM optimizers available in `cluster::pam()` (Maechler et al., 1999)

Optimizer	Description
K-medoids	Original PAM proposed in Kaufmann and Rousseeuw (1987)
PAM1	Optimizations proposed in Reynolds et al. (2006)
PAM2	
FastPAM1	Optimizations proposed in Schubert and Rousseeuw (2019)
FastPAM2	
FastPAM3	
FasterPAM	Optimization proposed in Schubert and Rousseeuw (2021)

5.3.3 Vowel centroids

If the data contains F1 and F2 measures on a Bark scale, these can be clustered using the clustering techniques mentioned in the previous subsections. Alternatively, they can be analysed in a density plot and then submitted to a centroid-based clustering specifically designed for formant measures. The density plot is the result of two-dimensional kernel density estimation (KDE) using the `kde2d()` function from the MASS package (Ripley and Venables, 2009). This plot can be used to inspect the formant values in a two-dimensional acoustic space. Note that speech is produced three-dimensionally and the density plot is thus always some form of abstract representation along two spectral (not articulatory) dimensions. Nevertheless, the KDE on formant values might be useful to obtain an inventory-like interpretation of the vowels in a certain dataset (whether phonemic or allophonic depends on the data). It is also possible to inspect the way certain vowel contrasts cluster.

The KDE density values in the plot are shown along the F2 (x-axis) and F1 (y-axis) using a specified number of grid points. The formant values need to be in Bark scale to account for critical bandwidth perception (Zwicker, 1961). The `representation` tab (Section 4.6) can be used to convert formant values in Hertz to Bark. Note that all available F1 and F2 values are used, regardless of whether these are taken as singleton measure or time-series measure. Note that time-series measures might cause more variation for density estimation than singleton measures.

In the density plot, higher degrees of density are darker (maximum density: black), which essentially visualizes the areas in which most F1 and F2 values occur. These areas can be taken as *vowel centroids*. Note that areas of high density

do not automatically correspond to where vowel phonemes occur. The degree of matching between high density areas and phonemic vowel identity merely depends on the data under analysis and the way in which it was elicited (more notes on interpretation below).

The size of the density matrix can be set by specifying the number of rows and columns, i.e., the resolution of the density plot. A too high resolution increases the chances of finding acoustically similar vowel centroids, whereas a too low resolution increases the chances of missing acoustically similar vowel centroids. The resolution should be chosen in accordance with the extreme values for each formant. For example, an F2 ranging between 5 and 15 Bark and a resolution set to 20, will break up the plot x-axis in intervals of 0.5 Bark. Whether this interval is too narrow or too wide, depends on the position on the frequency scale and the expected bandwidth of the formants.

Centroids are calculated by first listing all cells in the density matrix for which all 8 neighbouring values are lower in density (density peaks). Given the goal of finding high-density (high-probability) values in the two-dimensional (acoustic) space, a cut-off point can be set which discards a proportion of density values sorted from low to high. A cut-off point of 0.75 thus ignores the lowest 75% of the density values. Setting the cut-off point controls the extent to which low-density values end up as vowel centroid candidates (high cut-off values), or the extent to which low-density values are overlooked in datasets for which the vowel counts are heavily skewed (low cut-off values). Thus higher cut-off points result in more reliable centroid candidates at the expense of their amount; high cut-off points lead to less centroid candidates.

The remaining list of density peaks (or: vowel centroid candidates) is sorted by the density values in decreasing order. Even though this sorted list already gives a rough indication of the most suitable vowel centroid candidates, the acoustic distances between them are not yet taken into account. Vowel inventories tend to be build up by principles of maximum acoustic contrast, referred to as *dispersion* (e.g., Liljencrants, Lindblom, and Lindblom, 1972), with the almost universal tendency for inventories of three or more vowels to occupy at least the ‘corners’ of the acoustic space: /i/, /u/ and /a/ (Disner, 1984).

To account for vowel spacing tendencies, two distance measures are taken for each vowel centroid candidate. First, the average acoustic distance (Euclidean distance between the pairs of F1 and F2 values) to all others is taken for each candidate. These distance values favor vowel centroids in the corners of the acoustic space. Second, the average Euclidean distance between each candidate and all other centroid candidates with higher density values is taken. For the centroid candidate with the highest density value (first in candidate list), the average distance to all other candidates

is (again) taken. This procedure takes into account dispersion, i.e., high-density candidates, once chosen, affect the remaining acoustic space. The distances (two per candidate) are then summed. The resulting distance values are thus informed by the density ranking and are calculated to reduce the likelihood that high density vowel centroid candidates which are (too) close to each other in the acoustic space both end up as highly suitable candidates. Both the density values and the summed distance values are then rescaled (between 0 and 1) and summed. The resulting value is taken as an index of vowel centroid likelihood (maximum 2), with higher values indicating better suitability (correlating with higher density *and* larger acoustic distances from higher density candidates). The resulting list of candidates is then sorted by this index (decreasing order).

An overlay of a chosen number of vowel centroids as red dots in the density plot can be produced by setting the number of clusters. The chosen number can maximally be set to the number of candidates found in the data using the procedure just described. Any lower number than the maximum selects the candidates in order of their index ranking (higher ranked ones first). The most important information regarding each vowel centroid candidate is provided as a table underneath the density plot and can be stored for reference (Section 6). The table can be inspected for the variation in density, distance and/or index values. If their distribution is clearly bimodal, the mode with the higher value might indicate more reliable vowel centroids and could be taken as a selection criterion.

Once vowel centroids are chosen, they can be set as cluster centers. Clustering is then done by assigning each observation to the nearest centroid (Euclidean distance). Note that time-series formant measures are summarized by the central tendency measure chosen in the plot settings (Section 5.5) in order to calculate the distance using a two-dimensional representation (a single F1 value and a single F2 value). Thereafter, cluster outputs can be inspected in the ways described in Section 5.6.

KDE is based on Bayesian probability and thus influenced by the frequency of certain (combinations of) F1 and F2 values. The density values obtained through KDE are thus inherently skewed with uneven distributions of vowels. When the goal is obtaining an inventory-like collection of vowel centroids, it is important to carefully interpret the density plot. Two main parameters are therefore given to guide the interpretation: the size of the density matrix (i.e., resolution) and a cut-off point for discarding low-density values. Likewise, the selection of vowel centroids from a list of candidates is driven by density values *and* acoustic distance. In this way, it is possible to obtain vowel centroids with a low density, if they are acoustically distinct from other centroids. And it is possible to discard cluttered vowel centroids with a low density. If skewed or incomplete vowel distributions are expected, more

precise configuration of the resolution and cut-off point are required in order to allow inventory-like interpretations. Formant data drawn from representative numbers of vowels facilitates the centroids resembling the vowels.

5.4 Number of clusters

Finding the ideal number of clusters is key in performing cluster analysis. Before deciding on the number of clusters, it is recommended for the researcher to have a theoretically motivated estimation of this number, i.e. before obtaining any result from the analysis. For example, if the aim is to find a basic set of different lexical tone contours from words, around four to six clusters could be sufficient to accurately capture the contrasts. However, if the aim is to find a broad set of all lexical tone contrasts in combination with phrase intonation patterns, e.g. 15 clusters could be the minimum number needed. One way of determining the number of clusters is to run several rounds of analysis, each time with an increasing number of clusters (Kaland, 2021, Section 2-4). This is particularly useful in an exploratory context, where only a rough estimation of the number of clusters can be made. It is also recommended to run the cluster analysis with a number of clusters that exceeds the hypothesized number. In this way, the researcher can reduce the risk of missing relevant contrast that were unexpected. The dendrogram (HAC only) provides some initial guidance in finding the ideal number of clusters. Obtaining the dendrogram for a given dataset does not require any prior decisions on the number of clusters. That is, the dendrogram remains as is when changing the number of clusters.

An indication of the accuracy of the number of clusters can also be derived from plotting the contours per cluster, as further explained below. Apart from the general guidelines outlined here, there are statistical methods to obtain an estimation of the ideal number of clusters in an analysis, which are left for the user to explore (e.g. Charrad et al., 2014).

5.5 Plot settings

The ‘Clustering’ tab offer basic settings for the way the cluster plots are displayed. The default central tendency measure by which the clusters are summarized is the mean and standard deviation. Alternatively, the median can be chosen as central tendency measure with the coverage probability (0.95) of the outer quantiles (see package `Hmisc`). The median values and their coverage probability help to deal with and understand variation and outlying values in a cluster. Note that choosing either

central tendency measure is a plot setting and does not affect the way in which clusters are formed. The setting does affect the way all clustering variables are summarized, and does affect the computation of cluster prototypes (Section 5.9).

The number of columns for the cluster plot can be set, which determines how many cluster panels are displayed horizontally (one panel per cluster). This number, depending on the number of clusters, determines width of the panels (height is set to 400 pixels per row/panel).

It is furthermore possible to set which time-series variable should be represented on the y-axis. When multiple time-series variables are available in the data, a second y-axis (on the right of the plot) can be chosen. In this way, multiple representations can be visually inspected for how they compare among each other, independent of whether they were actually used as a variable for clustering or not. The second y-axis furthermore offers the option to plot the prototypical contours as calculated for each cluster (Section 5.9).

Note that the plot settings apply to displaying the plot in the application. Saving the plot takes the width and height as set in the application's general settings (Section 7).

5.6 Cluster outputs

After each clustering, a plot and tables are generated. HAC additionally provides a dendrogram for initial inspection of the results. The outputs are discussed in the following.

5.6.1 Dendrogram (HAC only)

HAC provides the researcher with a dendrogram, a tree-structure showing the outcome of each merge. The dendrogram provides a first visualisation of the clustering process. On the basis of the dendrogram the researcher can decide on the amount of clusters. The dendrogram also provides insight in the scale of the f0 differences between the clusters. Since the largest numerical differences between clusters are found at the top of the dendrogram, an initial analysis might reveal only the differences involving a larger f0 range. In such an initial outcome, the dendrogram is likely to show asymmetry. This asymmetry is the result of late adjoining of major f0 excursions (e.g. large boundary tones) with a cluster containing all smaller scale (e.g. phrase-internal) f0 excursions. Asymmetry can be avoided to some extent by choosing a speaker correction method that accounts for outliers. Initial clustering outcomes thus provide insight into contour differences of the largest scale. Smaller

scale differences, such as f0 peak height, are more likely to be successfully revealed when increasing the number of clusters. However, large numbers of clusters might result in clusters consisting of few contours, compromising the prototypicality of these contours. To accurately reveal small scale f0 differences, it is recommended to analyse a more homogeneous (controlled) dataset, or a subset of the data leaving out large-scale f0 excursions.

5.6.2 Plot

The plot shows the average values of the time-series variable(s) that were used for clustering (y) for each measurement point (x). The grey band around the values represents the standard deviations for the means or 0.95 coverage probability for the medians. A panel is displayed for each cluster, with the cluster number and number of observations in that cluster (n) given in the heading. If duration is taken into account as a clustering variable, the panel headings also display the mean/median duration (d) per cluster.

It is plausible that some of the clusters in the plot show highly similar contours. This is to be expected with a (recommended) high number of clusters for initial analysis. It is recommended to not solely rely on the plot of mean contours to inspect the potential similarity between two clusters. It might not be immediately clear why two similar contours end up in separate clusters. Inspection of the actual acoustic differences between the contours is recommended in such a case. This can be done by reading the f0 values from the plot. These values should be compared to more obvious differences between other clusters obtained from the same analysis. For example, rising phrase final boundary tones could have a large f0 range and therefore be visually easy to detect. This type of “landmark” in the contour can provide an indication of the scale of the differences between clusters. Thus, visually similar contours in separate clusters might also be the result of small numerical differences which could be more accurately clustered after subsetting. Decreasing the number of assumed clusters could also be considered as a means to obtain visually more distinguished contours (an example of such a consideration is given in Kaland, 2021, Section 3). However, this method makes the cluster analysis more course-grained, introducing a higher risk of overlooking relevant contour differences.

5.6.3 Table

The summary table shows the number of observations per cluster, the percentage of the number of observations in the cluster relative to all observations, and for each

clustering variable the mean standard error per cluster. The mean standard error per cluster is calculated by adding up the standard errors for each measurement point and divide the outcome by the number of measurement points. A tentative indication (flagging) is given of whether a cluster should be treated with caution. The flagging is given by a * based on the data in the table. That is, caution is advised when a cluster contains only one contour or when the mean standard error is more than two times the median of the mean standard errors of all clusters. The rationale behind the latter threshold is to obtain a single criterion that can be applied to f0 values regardless of the speaker correction method or number of clusters chosen. In particular, standard errors might be affected by some of these correction methods. By taking the median as center value, the distance from zero (no deviation) is known. Thus, mean standard error values that lie further away from median than zero does, are advised to be treated with caution. The criterion is a rough estimate of how deviant the mean contour in a particular cluster is.

Individual inspection is still advised to be able to quantify the standard errors on a f0 scale. Therefore, the inspection should be done before applying speaker correction. For example, a single cluster with a mean standard error of 10 Hz might already be enough to overlook potentially meaningful differences between the contours. That is, the f0 range for human speech can be taken as 75 to 500 Hz (standard setting in Praat; Boersma and Weenink, 2025). Thus, allowing for an average deviation of up to 10 Hz for the entire contour corresponds to a semitone difference that lies between 0.3 (480-500 Hz) and 2.2 (75-85 Hz). It has been shown that in Dutch, for example, 1.5 ST excursions (locally!) can be enough to perceive a linguistically meaningful prominence shift (Rietveld and Gussenhoven, 1985). Although the generalizability of this finding to other languages is open to further research, it is crucial to note that within cluster variance of 10 Hz might not fully avoid overlooking important f0 movements in the contour. The double-median threshold for the mean standard errors could therefore be too crude without consideration of the f0 scale. While smaller standard errors are generally preferred, increasing the number of assumed clusters can help to gain insight into this type of variability and is recommended prior to any subsetting procedures. The standard errors can thus be taken as an indication of how well the contours fit in the cluster. As a rule of thumb: if homogeneously sized clusters with a low mean standard error can be achieved with subsetting applied only to discard erroneous cases, the analysis is likely to provide an optimal outcome.

A contingency table can be generated after each clustering. The contingency table generates one row per cluster and columns for each level of a chosen variable. In this way, the contingency table may act as first interpretation of the clustering with regard to the variables that are relevant for the research question. The variable

can be copied/extracted from an existing character column (i.e., ‘interval_label’). The extraction steps are similar to the ones explained for mapping the ‘speaker’ column (Section 4.4):

- 1) select the column containing the variable
- 2) select the separator character (if any)
- 3) select the column with the relevant levels of the variable after separation.

Note that each unique character string found in step 3 will become a column in the contingency table.

5.7 Subsetting

A procedure for subsetting the data is provided in the application (tab ‘Clustering’, subtab ‘Table’). Clicking ‘Remove clusters’ provides an option to eliminate the observations in specific clusters. Note that this is irreversible within one session, i.e., data needs to be measured or uploaded again in order to have the complete dataset. The application automatically lists the flagged clusters as candidates for removal.

The removal procedure (henceforth ‘subsetting’) has two purposes. Its initial use can be applied to a high number of clusters (Kaland, 2021, Section 3 and 4 for examples) in order to remove erroneous or outlying contours. For this purpose, a high number of clusters (e.g., 25 or more) is recommended to obtain small sized clusters. If the data consists of erroneous or outlying contours, these will then be revealed in small sized clusters, potentially single-contour clusters. Thus, when removing small sized clusters, the risk of discarding error-free contours remains low. This way of “pruning” the data can be done using the automatic flagging of erroneous/outlying contours (Section 5.6.3).

A second purpose of the subsetting is to “zoom in” into a specific subset of the data, typically after initial round(s) of clustering. For example, initial clustering could reveal a small number of rising contours among an overall majority of falling contours. It could be useful to separate the rises and falls in subsets and perform subsequent clustering on either subset. This has the advantage of revealing smaller scale differences between the contours (e.g. different types of rises or falls), without their differences being affected by contours of the other category. Such an application of the subsetting procedure is particularly useful when there is clear indication or supportive evidence from distinguishing these two types as categories.

A general word of caution should be given here, as subsetting essentially ignores (potentially large) parts of the collected data. This procedure compromises

the representativeness of the empirical investigation and re-introduces the risk of giving researchers’ intuitions a decisive role in the analysis-process. Although these disadvantages cannot be entirely avoided, it is crucial to keep the ultimate goal of the investigation in mind. The procedures outlined here are designed to reveal prototypical contours, for which some deviant instances can be naturally expected in spontaneous speech (see further discussion in Kaland, 2021, Section 3).

5.8 Evaluation

The application provides an additional interface to evaluate the ideal number of clusters (tab ‘Evaluation’). Two methods are implemented; one based on information cost (Shannon, 1948) and minimum description length (MDL, Rissanen, 1978) as described in Kaland and Ellison (2023), and one based on within and between cluster variance. In case multiple clustering variables were chosen, the evaluation is based on the first (preventing the treatment of different scales as identical).

The first evaluation method computes information cost of three aspects of the cluster analysis; 1) the cost of specifying the clusters, which is expected to increase with more clusters, 2) the probability of a contour being in a specific cluster, given what the prototype (mean) of that cluster is, which is likely to decrease with more clusters, and 3) the cost of specifying each contour within a certain cluster, this is likely to increase with more clusters, depending on how well an individual contour fits the cluster (see details and demonstration in Kaland and Ellison, 2023). The summed outcome is then taken as a single evaluative information cost measure for one round of cluster analysis (one ‘clustering’).

When applied over a range of clusterings, e.g. starting with two clusters up to ten clusters, the evaluation measures for each clustering round are likely to produce a U shaped curve when the number of clusters are plotted on the x-axis and the information cost measure on the y-axis. The user is required to specify the range of cluster rounds and a bending factor (measurement point dependency value). The former indicates the number of clusters that should be assumed for the first and the last round of cluster analysis that should be evaluated. The evaluation then runs those rounds and all rounds with intermediate numbers of clusters. The dependency value indicates how dependent adjacent measurement points are, with higher values corresponding to higher degrees of dependency. Note that this is a way to take into account the length of the unit of analysis, which could range from (portions of) syllables (high dependency) to entire utterances (low dependency). The bending factor value can be manually set (with decimals, if needed). This might be needed

to obtain a clearer U shaped bend in the plotted curve, but should be informed by the unit of analysis and the absolute time-gap between measurement points. When analysing a dataset for which time-normalization has been done outside the provided Praat- or R-scripts, the bending factor needs to be adjusted accordingly.

The MDL evaluation procedure generates a plot with the information cost for each clustering round. The clustering round with the lowest information cost is the MDL and could be taken as the ideal number of clusters for the dataset under analysis. Note that all evaluation is based on the specific dataset under analysis and does therefore not directly allow generalization for the contour inventory of that specific language.

The second evaluation method is based on within and between cluster variation. It is expected that within cluster variance of the f0 contours decreases with more clusters, because with more clusters the individual contour within a certain cluster are more alike than with less clusters. In addition, between cluster variance is expected to increase with more clusters as the mean f0 contour in a certain cluster will differ more from the mean f0 contours in other clusters (i.e. more varying mean contours with more clusters). Note that these principles do not necessarily apply when choosing centroid-based linkage criteria (inversion), see Table 5.2.

Within cluster variance is computed by taking the standard deviation of each measurement point iteratively for all measurement points in a cluster. That is, starting with cluster 1, the standard deviation is taken from all the first measurement points in that cluster, then the standard deviation is taken from all the second measurement points in that cluster (etc.), until the final measurement point in that cluster. Then, the mean of all the standard deviations is taken to represent the variance in that cluster, after which the same procedure is repeated for the remaining cluster(s). Thereafter, all the mean standard deviations (one per cluster) are averaged again to represent within cluster variance for one clustering.

Between cluster variance is computed by taking the mean f0 value of a measurement point across clusters. That is, starting with measurement point 1, their mean value in each cluster is taken iteratively for all clusters. Then, the absolute difference between the maximum and the minimum of these mean values is taken to represent the between cluster variance of that particular measurement point. This procedure is repeated for all measurement points and the single value representing the between cluster variance of a particular clustering is the mean of all these difference measures.

In order to make the within and between cluster variance comparable, the values are scaled between 0 and 1. The scaling is done for each evaluation round. For example, if an evaluation round is set to compare clusterings ranging from 2 to 10 clusters (9 clusterings), the resulting 9 values representing within cluster variance as

well the 9 values representing the between cluster variance are scaled.

Theoretically, the optimum clustering is the one for which the lowest within cluster variance and the highest between cluster variance is observed. The optimum number of clusters will thus lie behind the cross-over point of both variances. Depending on the curvature of the variances a clear optimum might be observed, which is reached for the number of clusters before the distance between the two variances no longer or only minimally increases. Note that the cluster variance evaluation method is sensitive to the range of clusterings taken into account. That is, the scaled variance might vary to some extent depending on the range of clusterings. It is recommended to observe the relative gain or loss in the curves representing within/between cluster variance. If a ceiling or floor effect can be observed, this could be taken as an indication that extending the maximum of the selected range is no longer needed. Rather, if the ceiling or floor is stretched over multiple clustering rounds at the high end of the range, the optimum lies well before the clustering round with the highest number of clusters. It is recommended to have an informed idea about the maximum number of clusters to test in the evaluation.

The two evaluation methods do not need to show corresponding results. It is important to note that the approaches have different backgrounds and use therefore different ways to express the optimal number of clusters. One method seeks the optimum in terms of informativeness of describing the dataset. The other method provides insight into the clustering process and how the contour variation is distributed within or between clusters. That is to say that they can both provide useful perspectives on what the number of clusters to choose *could* be.

5.9 Prototypes

After each clustering, prototypical contours in the data are marked with an ‘x’ in the column ‘proto’ in ‘df.l’. The listed contours are summarized in a table for convenience in the ‘Prototypes’ tab. If acoustic measurements were taken in the application (Section 4.2 and 4.3), the prototype table also provides play buttons to listen to each of the prototypes. The sampled prototypes are cut from the original file and stored temporarily in the ‘www’ folder in the folder from which the application is run. Prototypical contours can also be plotted on a second y-axis in the plot (Section 5.5).

The prototypes are calculated as follows. For all observations in a given cluster, the Euclidean distance to the central tendency of that cluster is calculated (mean/-median, depending on what is chosen in the plot settings, Section 5.5). The observa-

tions with the smallest distance to the central one is taken as the most prototypical in a cluster. All acoustic variables chosen for clustering are also taken into account for prototype calculation by scaling the distances for each variable and choosing the prototype as the observation with the smallest sum of all scaled distances.

6. Data conventions

The following sections describe the main data objects stored in R’s working memory by the names with which they are stored. This explanation provides (advanced) users with a basic understanding of how the application handles the data. All objects with a name starting with ‘df.’ refer to data tables (package `data.table`, Barrett et al., 2006). It is possible to directly access these objects from the R Global Environment (see Section 7.6).

6.1 df.u

This is the data as uploaded by the user and read by `data.table::fread()`. This data is displayed in the tab ‘Data (upload)’ immediately after uploading and before mapping the columns to the required ones. If uploaded data is not correctly displayed, either change the way the data is stored or specify additional options in the syntax of the `fread()` function.

6.2 df.g

This is the data as read from the textgrids by `readtextgrid::read_textgrid()` (Mahr, 2020). The data provides all information in the textgrids as a data table and can be viewed in the ‘Data (TextGrids)’ tab. This tab offers a way to check whether the textgrids are read correctly.

6.3 df.l

This is the main data with which the application works. It is generated on the basis of the uploaded (‘df.u’) or read (‘df.g’) data and has the long format. This means

that each row represents a single measurement point. Table 4.1 provides an overview of the columns that are minimally needed in this data.

All columns that refer to variables for clustering have a specific template for their naming. The names are concatenated character strings following the structure in Table 6.1.

Table 6.1: Clustering variable column naming conventions in ‘df.l’. Concatenation of strings from left to right, one string taken per column in the table, with optional strings in brackets. N.B. octave-median rescaling, derivation and fPCA are only available for f0. Standardization is available for all cues.

default	cue	scale	[spk.corr]	[derivation]	[fPCA]
		.Hz	.oMe		
	.f0	.ERB	.std	.d{n}	.PC{n}
		.ST			
cc	.int	.dB	.std		
	.dur	.s	.std		
	.f{n}	.Hz	.std		
		.Bk			

For example, `cc.f0.ERB.oMe.PC2` is the name of the column that contains the PC2 values taken from octave-median rescaled ERB values of fundamental frequency. And `cc.f1.Bk` is the name of the column that contains the Bark values of the first formant. And `cc.dur.std` is the name of the column with standardized duration values. `cc.f0.Hz.d3` is the name of the column that contains the Hertz values of the third derivative (jerk) of f0.

6.3.1 Singleton vs. time-series

In the long data format each measurement point appears on a new row, which is mainly the case to cater for time-series variables such as f0 and intensity. This is because time-series measures represent the interval by the chosen number of measurement points. Duration values and principal component values are singleton measures in the sense that they represent the interval by a single value. Singleton measures are written to the data at stepnumber 1 (the other measurement points are NA). Formant values can be represented as time-series or as singleton measures (Section 4.3.8).

Clustering with singleton measures restricts the use of distance metrics. That is, only distance metrics based on Euclidean distance are available. Other distance metrics are exclusively applicable to time-series measures.

6.3.2 Decimals

All numeric values in ‘df.l’ have a maximum of three decimals, in order to avoid bloated datafiles and reduce processing speed.

6.4 df.e

This data is obtained from the evaluation procedure and lists for each clustering round either an information cost value (MDL) or a combination of within and between cluster variance. This data is used to generate the evaluation plot and is displayed underneath it.

6.5 df.v

The vowel centroids candidates (Section 5.3.3) are stored with their density matrix row and a column index, density value, f1 and f2 values (Bark), summed Euclidean distances and the index values (rescaled density and distance values). These are displayed underneath the density plot and as data table named ‘df.v’.

7. Settings

A number of general settings is provided that apply to one session of running the application. They are discussed one by one in the following.

7.1 Audio conversion

In case the audio files have a sampling rate higher 32 kHz, a bit depth higher than 16 bits or are recorded in stereo, a warning is given. This is done to alert that audio conversion might speed up processing, in particular due to file size. In case the application handles the audio files well as they are, no conversion is necessary.

A common cause for audio reading problems in R are floating-point wave-files (IEEE float format), which have a bit depth of 32 or 64 bits (hence the warning described above). It is recommended to avoid the use of floating point wave files and convert them. The quality advantages of floating point wave files overshoots those needed for the kind of speech processing done in the application.

A conversion option is offered within the application to obtain (usually) smaller, good quality and readable audio files. The converter requires a path to the audio files that need conversion and a directory name to store the converted files in. The default output is set to PCM wave, bit depth: 16 bits, sampling rate: 16 kHz, mono. This format facilitates processing in the application and warrants sufficient quality for recorded speech (e.g., Jayant and Noll, 1984). When processing large amounts of data, audio conversion can also speed up processing.

7.2 File upload limit

This sets the limit for uploading files to the application in megabytes. Depending on the computer, the way in which the application is run, and the size of the data, this setting provides a way to prevent the application from being slow due to the upload

of large datasets. Note that with each addition of an acoustic cue representation (Section 4.6), additional values are written to the data and will make the data object larger in size after uploading. Adding a representation after a file has been uploaded is not affected by the upload limit. Therefore, it could be more important to start with a small dataset when uploading.

7.3 Number of rows to display

This setting specifies how many rows should be displayed when viewing data (any of the sets described in Section 6. This number is by default limited to 50 as to prevent large datasets requiring long rendering times.

7.4 Saving

The application provides options to save the logfile, the data, the cluster plot, the summary table, the contingency table, the prototype table and the evaluation plot/table. Individual ‘Save’ buttons are available for each. Alternatively, ‘Save this’ can be clicked to save all available output at once in a folder called ‘saved’ (in the directory from which the application is run). Additional saving settings are provided in the application settings. These include setting the width and height in pixels for all plots that can be saved. It is furthermore possible to generate a compressed file (.zip) with for all outputs saved through the ‘Save this’ button. The compressed file is given a date and number of clusters in the filename for future reference.

Note that the data in long format is saved such that it can be re-uploaded to the application in a future session (see also Section 6).

7.5 Line aesthetics

It might be useful to distinguish the lines in the plot visually, in particular when working with multiple Y-variables (see Section 5.5). This can be done by setting the line type (solid or dashed) and line color (any hex code).

7.6 Keeping objects

In order to keep the objects that were generated by the application in the Global Environment of R, tick this option. Keeping objects is useful for debugging purposes

or for continuing to work with the data after the application was closed. See Section 6 for an overview of the relevant objects. Keeping objects in the Global Environment lasts as long as the R-session and still requires saving them manually if this is desired. By default, the objects generated by the application are removed upon closing.

7.7 Garbage collection

Runs R's garbage collection function `gc()`. This function is only needed if the application is slow, in which case the effects might be minimal. Garbage collection happens automatically in R regardless of manually running this function.

7.8 Logfile

The application automatically writes a lines of text to a logfile for each relevant step in the workflow. The logfile provides all information needed to reproduce the specific steps in a session. It is highly recommended to save the logfile each time any of the cluster outputs are saved.

8. Software statements

8.1 Data protection

Even though it is possible to run the application in a browser, the application is not hosted on a web-server and needs to be downloaded to run locally on the user's computer. Any data (read folder or uploaded datafile) will therefore be handled within the R environment on the user's local computer. The application has no functionality that transfers or shares the data to any other location than specified by the R environment, which defaults to the user's local computer (i.e., local hard disk, unless specified differently in R or R Studio).

By default, all objects in the R environment are removed upon closing the application. Objects may remain loaded in the R environment after the application crashes. It is furthermore possible for the user to specify that specific objects remain available in the R environment after closing the application (Section 7.6). Importantly, all operations carried out by the application can be inspected or altered, because the application follows the principles of free and open-source software (FOSS; e.g., Fortunato and Galassi, 2021). All software code referred to as 'the application' in this manual is contained in a single .R file, which can be downloaded from: <https://constantijnkaland.github.io/contourclustering/#download>.

8.1.1 Citation

Please cite Contour Clustering using the following references:

Kaland, C. (2021). Contour clustering: A field-data-driven approach for documenting and analysing prototypical f0 contours. *Journal of the International Phonetic Association*, 53(1), 159–188. <https://doi.org/10.1017/S0025100321000049>

Kaland, C. (2026). Contour clustering [Computer software]. Version 09-2026.

<https://constantijnkaland.github.io/contourclustering/>

Later work following Kaland (2021) investigates specific aspects of the application of clustering to intonation and prosody, and is given in Table 8.1.

Table 8.1: Overview of literature on cluster analysis applied to prosodic phenomena.

Study	Content	Acoustic cue(s)
Kaland et al., 2026	Word-level use of f0 in Yali	f0
Kaland, 2025	F0 derivatives in clustering	f0, duration, intensity
Kaland, Steffman, and Cole, 2024	HAC compared to K-means clustering	f0
Kaland and Grice, 2024	Cluster analysis combined with random forests	f0
Kaland and Ellison, 2023	Evaluating number of clusters (HAC)	f0
Kaland, 2023	Distance metrics compared to perceived similarity	f0
Kaland, 2021	Introduction of the application and theoretical motivation	f0

An overview of literature in which the application was used to study prosodic phenomena in a variety of languages can be found here:

<https://constantijnkaland.github.io/contourclustering/#languages>.

8.2 License

The application is published under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International license (CC BY-NC-SA 4.0). The full legal text is provided as Appendix A in Section 9. The following provides a summary of the terms from <https://creativecommons.org/licenses/by-nc-sa/4.0/>.

CC BY-NC-SA 4.0	
You are free to:	
Share	- copy and redistribute the material in any medium or format
Adapt	- remix, transform, and build upon the material
The licensor cannot revoke these freedoms as long as you follow the license terms.	
Under the following terms:	

- | | | |
|----------------------------|---|---|
| Attribution | - | You must give appropriate credit, provide a link to the license, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use. |
| NonCommercial | - | You may not use the material for commercial purposes. |
| ShareAlike | - | If you remix, transform, or build upon the material, you must distribute your contributions under the same license as the original. |
| No additional restrictions | - | You may not apply legal terms or technological measures that legally restrict others from doing anything the license permits. |

This deed highlights only some of the key features and terms of the actual license. It is not a license and has no legal value. You should carefully review all of the terms and conditions of the actual license before using the licensed material.

9. Appendix A: License

Attribution-NonCommercial-ShareAlike 4.0 International

=====

Creative Commons Corporation ("Creative Commons") is not a law firm and does not provide legal services or legal advice. Distribution of Creative Commons public licenses does not create a lawyer-client or other relationship. Creative Commons makes its licenses and related information available on an "as-is" basis. Creative Commons gives no warranties regarding its licenses, any material licensed under their terms and conditions, or any related information. Creative Commons disclaims all liability for damages resulting from their use to the fullest extent possible.

Using Creative Commons Public Licenses

Creative Commons public licenses provide a standard set of terms and conditions that creators and other rights holders may use to share original works of authorship and other material subject to copyright and certain other rights specified in the public license below. The following considerations are for informational purposes only, are not exhaustive, and do not form part of our licenses.

Considerations for licensors: Our public licenses are intended for use by those authorized to give the public permission to use material in ways otherwise restricted by copyright and certain other rights. Our licenses are irrevocable. Licensors should read and understand the terms and conditions of the license they choose before applying it. Licensors should also secure all rights necessary before applying our licenses so that the public can reuse the material as expected. Licensors should clearly mark any material not subject to the license. This includes other CC-licensed material, or material used under an exception or limitation to copyright. More considerations for licensors:

wiki.creativecommons.org/Considerations_for_licensors

Considerations for the public: By using one of our public licenses, a licensor grants the public permission to use the licensed material under specified terms and conditions. If the licensor's permission is not necessary for any reason--for example, because of any applicable exception or limitation to copyright--then that use is not regulated by the license. Our licenses grant only permissions under copyright and certain other rights that a licensor has authority to grant. Use of the licensed material may still be restricted for other reasons, including because others have copyright or other rights in the material. A licensor may make special requests, such as asking that all changes be marked or described. Although not required by our licenses, you are encouraged to respect those requests where reasonable. More considerations for the public: wiki.creativecommons.org/Considerations_for_licensees

=====

Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International Public License

By exercising the Licensed Rights (defined below), You accept and agree to be bound by the terms and conditions of this Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International Public License ("Public License"). To the extent this Public License may be interpreted as a contract, You are granted the Licensed Rights in consideration of Your acceptance of these terms and conditions, and the Licensor grants You such rights in consideration of benefits the Licensor receives from making the Licensed Material available under these terms and conditions.

Section 1 -- Definitions.

a. Adapted Material means material subject to Copyright and Similar Rights that is derived from or based upon the Licensed Material and in which the Licensed Material is translated, altered, arranged, transformed, or otherwise modified in a manner requiring permission under the Copyright and Similar Rights held by the Licensor. For purposes of this Public License, where the Licensed Material is a musical work, performance, or sound recording, Adapted Material is always produced where the Licensed Material is synched in timed relation with a moving image.

b. Adapter's License means the license You apply to Your Copyright and Similar Rights in Your contributions to Adapted Material in accordance with the terms and conditions of this Public License.

- c. BY-NC-SA Compatible License means a license listed at creativecommons.org/compatiblelicenses, approved by Creative Commons as essentially the equivalent of this Public License.
- d. Copyright and Similar Rights means copyright and/or similar rights closely related to copyright including, without limitation, performance, broadcast, sound recording, and Sui Generis Database Rights, without regard to how the rights are labeled or categorized. For purposes of this Public License, the rights specified in Section 2(b)(1)-(2) are not Copyright and Similar Rights.
- e. Effective Technological Measures means those measures that, in the absence of proper authority, may not be circumvented under laws fulfilling obligations under Article 11 of the WIPO Copyright Treaty adopted on December 20, 1996, and/or similar international agreements.
- f. Exceptions and Limitations means fair use, fair dealing, and/or any other exception or limitation to Copyright and Similar Rights that applies to Your use of the Licensed Material.
- g. License Elements means the license attributes listed in the name of a Creative Commons Public License. The License Elements of this Public License are Attribution, NonCommercial, and ShareAlike.
- h. Licensed Material means the artistic or literary work, database, or other material to which the Licensors applied this Public License.
- i. Licensed Rights means the rights granted to You subject to the terms and conditions of this Public License, which are limited to all Copyright and Similar Rights that apply to Your use of the Licensed Material and that the Licensors has authority to license.
- j. Licensors means the individual(s) or entity(ies) granting rights under this Public License.
- k. NonCommercial means not primarily intended for or directed towards commercial advantage or monetary compensation. For purposes of this Public License, the exchange of the Licensed Material for other material subject to Copyright and Similar Rights by digital file-sharing or similar means is NonCommercial provided there is no payment of monetary compensation in connection with the exchange.
- l. Share means to provide material to the public by any means or process that requires permission under the Licensed Rights, such as reproduction, public display, public performance, distribution, dissemination,

communication, or importation, and to make material available to the public including in ways that members of the public may access the material from a place and at a time individually chosen by them.

m. Sui Generis Database Rights means rights other than copyright resulting from Directive 96/9/EC of the European Parliament and of the Council of 11 March 1996 on the legal protection of databases, as amended and/or succeeded, as well as other essentially equivalent rights anywhere in the world.

n. You means the individual or entity exercising the Licensed Rights under this Public License. Your has a corresponding meaning.

Section 2 -- Scope.

a. License grant.

1. Subject to the terms and conditions of this Public License, the Licensor hereby grants You a worldwide, royalty-free, non-sublicensable, non-exclusive, irrevocable license to exercise the Licensed Rights in the Licensed Material to:

a. reproduce and Share the Licensed Material, in whole or in part, for NonCommercial purposes only; and

b. produce, reproduce, and Share Adapted Material for NonCommercial purposes only.

2. Exceptions and Limitations. For the avoidance of doubt, where Exceptions and Limitations apply to Your use, this Public License does not apply, and You do not need to comply with its terms and conditions.

3. Term. The term of this Public License is specified in Section 6(a).

4. Media and formats; technical modifications allowed. The Licensor authorizes You to exercise the Licensed Rights in all media and formats whether now known or hereafter created, and to make technical modifications necessary to do so. The Licensor waives and/or agrees not to assert any right or authority to forbid You from making technical modifications necessary to exercise the Licensed Rights, including technical modifications necessary to circumvent Effective Technological Measures. For purposes of this Public License, simply making modifications authorized by this Section 2(a) (4) never produces Adapted Material.

5. Downstream recipients.

a. Offer from the Licensor -- Licensed Material. Every recipient of the Licensed Material automatically receives an offer from the Licensor to exercise the Licensed Rights under the terms and conditions of this Public License.

b. Additional offer from the Licensor -- Adapted Material. Every recipient of Adapted Material from You automatically receives an offer from the Licensor to exercise the Licensed Rights in the Adapted Material under the conditions of the Adapter's License You apply.

c. No downstream restrictions. You may not offer or impose any additional or different terms or conditions on, or apply any Effective Technological Measures to, the Licensed Material if doing so restricts exercise of the Licensed Rights by any recipient of the Licensed Material.

6. No endorsement. Nothing in this Public License constitutes or may be construed as permission to assert or imply that You are, or that Your use of the Licensed Material is, connected with, or sponsored, endorsed, or granted official status by, the Licensor or others designated to receive attribution as provided in Section 3(a)(1)(A)(i).

b. Other rights.

1. Moral rights, such as the right of integrity, are not licensed under this Public License, nor are publicity, privacy, and/or other similar personality rights; however, to the extent possible, the Licensor waives and/or agrees not to assert any such rights held by the Licensor to the limited extent necessary to allow You to exercise the Licensed Rights, but not otherwise.

2. Patent and trademark rights are not licensed under this Public License.

3. To the extent possible, the Licensor waives any right to collect royalties from You for the exercise of the Licensed Rights, whether directly or through a collecting society under any voluntary or waivable statutory or compulsory licensing scheme. In all other cases the Licensor expressly reserves any right to collect such royalties, including when the Licensed Material is used other than for NonCommercial purposes.

Section 3 -- License Conditions.

Your exercise of the Licensed Rights is expressly made subject to the following conditions.

a. Attribution.

1. If You Share the Licensed Material (including in modified form), You must:

a. retain the following if it is supplied by the Licensor with the Licensed Material:

i. identification of the creator(s) of the Licensed Material and any others designated to receive attribution, in any reasonable manner requested by the Licensor (including by pseudonym if designated);

ii. a copyright notice;

iii. a notice that refers to this Public License;

iv. a notice that refers to the disclaimer of warranties;

v. a URI or hyperlink to the Licensed Material to the extent reasonably practicable;

b. indicate if You modified the Licensed Material and retain an indication of any previous modifications; and

c. indicate the Licensed Material is licensed under this Public License, and include the text of, or the URI or hyperlink to, this Public License.

2. You may satisfy the conditions in Section 3(a)(1) in any reasonable manner based on the medium, means, and context in which You Share the Licensed Material. For example, it may be reasonable to satisfy the conditions by providing a URI or hyperlink to a resource that includes the required information.

3. If requested by the Licensor, You must remove any of the information required by Section 3(a)(1)(A) to the extent reasonably practicable.

b. ShareAlike.

In addition to the conditions in Section 3(a), if You Share Adapted Material You produce, the following conditions also apply.

1. The Adapter's License You apply must be a Creative Commons license with the same License Elements, this version or later, or a BY-NC-SA Compatible License.

2. You must include the text of, or the URI or hyperlink to, the Adapter's License You apply. You may satisfy this condition in any reasonable manner based on the medium, means, and context in which You Share Adapted Material.

3. You may not offer or impose any additional or different terms or conditions on, or apply any Effective Technological Measures to, Adapted Material that restrict exercise of the rights granted under the Adapter's License You apply.

Section 4 -- Sui Generis Database Rights.

Where the Licensed Rights include Sui Generis Database Rights that apply to Your use of the Licensed Material:

- a. for the avoidance of doubt, Section 2(a)(1) grants You the right to extract, reuse, reproduce, and Share all or a substantial portion of the contents of the database for NonCommercial purposes only;
- b. if You include all or a substantial portion of the database contents in a database in which You have Sui Generis Database Rights, then the database in which You have Sui Generis Database Rights (but not its individual contents) is Adapted Material, including for purposes of Section 3(b); and
- c. You must comply with the conditions in Section 3(a) if You Share all or a substantial portion of the contents of the database.

For the avoidance of doubt, this Section 4 supplements and does not replace Your obligations under this Public License where the Licensed Rights include other Copyright and Similar Rights.

Section 5 -- Disclaimer of Warranties and Limitation of Liability.

a. UNLESS OTHERWISE SEPARATELY UNDERTAKEN BY THE LICENSOR, TO THE EXTENT POSSIBLE, THE LICENSOR OFFERS THE LICENSED MATERIAL AS-IS AND AS-AVAILABLE, AND MAKES NO REPRESENTATIONS OR WARRANTIES OF ANY KIND CONCERNING THE LICENSED MATERIAL, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHER. THIS INCLUDES, WITHOUT LIMITATION, WARRANTIES OF TITLE, MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, NON-INFRINGEMENT, ABSENCE OF LATENT OR OTHER DEFECTS, ACCURACY, OR THE PRESENCE OR ABSENCE OF ERRORS, WHETHER OR NOT KNOWN OR DISCOVERABLE. WHERE DISCLAIMERS OF WARRANTIES ARE NOT ALLOWED IN FULL OR IN PART, THIS DISCLAIMER MAY NOT APPLY TO YOU.

b. TO THE EXTENT POSSIBLE, IN NO EVENT WILL THE LICENSOR BE LIABLE TO YOU ON ANY LEGAL THEORY (INCLUDING, WITHOUT LIMITATION, NEGLIGENCE) OR OTHERWISE FOR ANY DIRECT, SPECIAL, INDIRECT, INCIDENTAL, CONSEQUENTIAL, PUNITIVE, EXEMPLARY, OR OTHER LOSSES, COSTS, EXPENSES, OR DAMAGES ARISING OUT OF THIS PUBLIC LICENSE OR USE OF THE LICENSED MATERIAL, EVEN IF THE LICENSOR HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH LOSSES, COSTS, EXPENSES, OR DAMAGES. WHERE A LIMITATION OF LIABILITY IS NOT ALLOWED IN FULL OR IN PART, THIS LIMITATION MAY NOT APPLY TO YOU.

c. The disclaimer of warranties and limitation of liability provided above shall be interpreted in a manner that, to the extent possible, most closely approximates an absolute disclaimer and waiver of all liability.

Section 6 -- Term and Termination.

a. This Public License applies for the term of the Copyright and Similar Rights licensed here. However, if You fail to comply with this Public License, then Your rights under this Public License terminate automatically.

b. Where Your right to use the Licensed Material has terminated under Section 6(a), it reinstates:

1. automatically as of the date the violation is cured, provided it is cured within 30 days of Your discovery of the violation; or
2. upon express reinstatement by the Licensor.

For the avoidance of doubt, this Section 6(b) does not affect any right the Licensor may have to seek remedies for Your violations of this Public License.

c. For the avoidance of doubt, the Licensor may also offer the Licensed Material under separate terms or conditions or stop distributing the Licensed Material at any time; however, doing so will not terminate this Public License.

d. Sections 1, 5, 6, 7, and 8 survive termination of this Public License.

Section 7 -- Other Terms and Conditions.

a. The Licensor shall not be bound by any additional or different terms or conditions communicated by You unless expressly agreed.

b. Any arrangements, understandings, or agreements regarding the Licensed Material not stated herein are separate from and independent of the terms and conditions of this Public License.

Section 8 -- Interpretation.

a. For the avoidance of doubt, this Public License does not, and shall not be interpreted to, reduce, limit, restrict, or impose conditions on any use of the Licensed Material that could lawfully be made without permission under this Public License.

b. To the extent possible, if any provision of this Public License is deemed unenforceable, it shall be automatically reformed to the minimum extent necessary to make it enforceable. If the provision cannot be reformed, it shall be severed from this Public License without affecting the enforceability of the remaining terms and conditions.

c. No term or condition of this Public License will be waived and no failure to comply consented to unless expressly agreed to by the Licensor.

d. Nothing in this Public License constitutes or may be interpreted as a limitation upon, or waiver of, any privileges and immunities that apply to the Licensor or You, including from the legal processes of any jurisdiction or authority.

=====
Creative Commons is not a party to its public licenses. Notwithstanding, Creative Commons may elect to apply one of its public licenses to material it publishes and in those instances will be considered the \Licensor." The text of the Creative Commons public licenses is dedicated to the public domain under the CCO Public Domain Dedication. Except for the limited purpose of indicating that material is shared under a Creative Commons public license or as otherwise permitted by the Creative Commons policies published at creativecommons.org/policies, Creative Commons does not authorize the use of the trademark "Creative Commons" or any other trademark or logo of Creative Commons without its prior written consent including, without limitation, in connection with any unauthorized modifications to any of its public licenses or any other arrangements, understandings, or agreements concerning use of licensed material. For the avoidance of doubt, this paragraph does not form part of the public licenses.

Creative Commons may be contacted at creativecommons.org.

10. References

- Barrett, T., Dowle, M., Srinivasan, A., Gorecki, J., Chirico, M., Hocking, T., & Schwendinger, B. (2006). Data.table: Extension of ‘data.frame’. <https://doi.org/10.32614/CRAN.package.data.table>
- Boersma, P., & Weenink, D. (2025). Praat: Doing Phonetics by Computer.
- Chang, W., Cheng, J., Allaire, J., Sievert, C., Schloerke, B., Xie, Y., Allen, J., McPherson, J., Dipert, A., & Borges, B. (2012). Shiny: Web Application Framework for R. <https://doi.org/10.32614/CRAN.package.shiny>
- Charrad, M., Ghazzali, N., Boiteau, V., & Niknafs, A. (2014). NbClust: An R Package for Determining the Relevant Number of Clusters in a Data Set. *Journal of Statistical Software*, 61(6). <https://doi.org/10.18637/jss.v061.i06>
- Culhane, K., & Kaland, C. (2025). Acoustic correlates of stress in Kupang Malay: A preliminary analysis. *The Third International Conference on Tone and Intonation*, 1–5. <https://doi.org/10.21437/TAI.2025-1>
- De Looze, C., & Hirst, D. (2014). The OMe (Octave-Median) scale: A natural scale for speech melody. *7th International Conference on Speech Prosody 2014*, 910–914. <https://doi.org/10.21437/SpeechProsody.2014-170>
- Disner, S. F. (1984). Insights on vowel spacing. In I. Maddieson (Ed.), *Patterns of sounds* (pp. 136–155). Cambridge University Press. Cambridge.
- Esling, P., & Agon, C. (2012). Time-series data mining. *ACM Computing Surveys*, 45(1), 1–34. <https://doi.org/10.1145/2379776.2379788>
- Fortunato, L., & Galassi, M. (2021). The case for free and open source software in research and scholarship. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 379(2197), rsta.2020.0079, 20200079. <https://doi.org/10.1098/rsta.2020.0079>
- Goldstein, J. L. (1973). An optimum processor theory for the central formation of the pitch of complex tones. *The Journal of the Acoustical Society of America*, 54(6), 1496–1516. <https://doi.org/10.1121/1.1914448>

- Gordon, M., & Ladefoged, P. (2001). Phonation types: A cross-linguistic overview. *Journal of Phonetics*, 29(4), 383–406. <https://doi.org/10.1006/jpho.2001.0147>
- Greenwood, D. D. (1961). Critical Bandwidth and the Frequency Coordinates of the Basilar Membrane. *The Journal of the Acoustical Society of America*, 33(10), 1344–1356. <https://doi.org/10.1121/1.1908437>
- Gubian, M., Torreira, F., & Boves, L. (2015). Using Functional Data Analysis for investigating multidimensional dynamic phonetic contrasts. *Journal of Phonetics*, 49, 16–40. <https://doi.org/10.1016/j.wocn.2014.10.001>
- Hermes, D. J. (1998). Measuring the Perceptual Similarity of Pitch Contours. *Journal of Speech, Language, and Hearing Research*, 41(1), 73–82. <https://doi.org/10.1044/jslhr.4101.73>
- Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679–688. <https://doi.org/10.1016/j.ijforecast.2006.03.001>
- Jayant, N. S., & Noll, P. (1984). *Digital coding of waveforms: Principles and applications to speech and video* (5. print). Prentice-Hall.
- Jun, S.-A., & Fletcher, J. (2014). Methodology of studying intonation: From data collection to data analysis. In S.-A. Jun (Ed.), *Prosodic Typology II* (pp. 493–519). Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199567300.003.0016>
- Kaland, C. (2019). Acoustic correlates of word stress in Papuan Malay. *Journal of Phonetics*, 74, 55–74. <https://doi.org/10.1016/j.wocn.2019.02.003>
- Kaland, C. (2021). Contour clustering: A field-data-driven approach for documenting and analysing prototypical f0 contours. *Journal of the International Phonetic Association*, 53(1), 159–188. <https://doi.org/10.1017/S0025100321000049>
- Kaland, C. (2023). Intonation contour similarity: F0 representations and distance measures compared to human perception in two languages. *The Journal of the Acoustical Society of America*, 154(1), 95–107. <https://doi.org/10.1121/10.0019850>
- Kaland, C. (2025). F0 derivatives in the classification of meaningful tonal movements. *Journal of Phonetics*, 113, 101454. <https://doi.org/10.1016/j.wocn.2025.101454>
- Kaland, C., & Ellison, T. M. (2023). Evaluating cluster analysis on f0 contours: An information theoretic approach on three languages. In R. Skarnitzl & J. Volín (Eds.), *Proceedings of the 20th International Congress of Phonetic Sciences* (pp. 3448–3452). Guarant International.

- Kaland, C., Gayler, K., Grice, M., & Himmelmann, N. P. (2026). A first exploration of word- and phrase-final f0 movements in spontaneous Yali. *Phonological Data and Analysis*, 8, e4. <https://doi.org/10.1017/S2642182825000030>
- Kaland, C., & Grice, M. (2024). Exploring and explaining variation in phrase-final f0 movements in spontaneous Papuan Malay. *Phonetica*, 81(3). <https://doi.org/10.1515/phon-2023-0031>
- Kaland, C., Steffman, J., & Cole, J. (2024). K-means and hierarchical clustering of f0 contours. *Interspeech 2024*, 1520–1524. <https://doi.org/10.21437/Interspeech.2024-181>
- Kaufman, L., & Rousseeuw, P. J. (Eds.). (1990). *Finding Groups in Data*. John Wiley & Sons, Inc. <https://doi.org/10.1002/9780470316801>
- Kaufmann, L., & Rousseeuw, P. (1987). Clustering by means of medoids. In Y. Dodge (Ed.), *Data Analysis based on the L1-Norm and Related Methods* (pp. 405–416). North Holland / Elsevier.
- Kisler, T., Reichel, U., & Schiel, F. (2017). Multilingual processing of speech via web services. *Computer Speech & Language*, 45, 326–347. <https://doi.org/10.1016/j.csl.2017.01.005>
- Liljencrants, J., Lindblom, B., & Lindblom, B. (1972). Numerical Simulation of Vowel Quality Systems: The Role of Perceptual Contrast. *Language*, 48(4), 839. <https://doi.org/10.2307/411991>
- Lindblom, B. (1986). Phonetic universals in vowel systems. In J. Ohala & J. Jaeger (Eds.), *Experimental Phonology* (pp. 13–44). Academic Press.
- Lobanov, B. M. (1971). Classification of Russian Vowels Spoken by Different Speakers. *The Journal of the Acoustical Society of America*, 49(2B), 606–608. <https://doi.org/10.1121/1.1912396>
- Maechler, M., Rousseeuw, P., Struyf, A., & Hubert, M. (1999). Cluster: "Finding Groups in Data": Cluster Analysis Extended Rousseeuw et al. <https://doi.org/10.32614/CRAN.package.cluster>
- Mahr, T. (2020). Readtextgrid: Read in a 'Praat' 'TextGrid' File. <https://doi.org/10.32614/CRAN.package.readtextgrid>
- Marchini, G., Steffman, J., & Kaland, C. (2025). Clustering methods in the analysis of intonation: The case of focus marking in Afro-Mexican Spanish. *The Third International Conference on Tone and Intonation*, 94–98. <https://doi.org/10.21437/TAI.2025-20>
- Max Planck Institute for Psycholinguistics. (2022). ELAN. <https://archive.mpi.nl/tla/elan>.

- McAuliffe, M., Socolof, M., Mihuc, S., Wagner, M., & Sonderegger, M. (2017). Montreal Forced Aligner: Trainable Text-Speech Alignment Using Kaldi. *Interspeech 2017*, 498–502. <https://doi.org/10.21437/Interspeech.2017-1386>
- Mixdorff, H., & Niebuhr, O. (2013). The influence of F0 contour continuity on prominence perception. *Interspeech 2013*, 230–234. <https://doi.org/10.21437/Interspeech.2013-73>
- Nespor, M., & Vogel, I. (2007). *Prosodic Phonology: With a New Foreword*. New York : Mouton de Gruyter.
- Noll, A., Stuefer, J., Klingler, N., Leykum, H., Lozo, C., Luttenberger, J., Pucher, M., & Schmid, C. (2019). Sound tools eXtended (stx) 5.0 — a powerful sound analysis tool optimized for speech. *Interspeech 2019*, 2370–2371.
- R Core Team. (2025). R: The R project for statistical computing. Version 4.2.1.
- R Studio Team. (2025). RStudio: Integrated Development for R.
- Ramsay, J. (2003). Fda: Functional Data Analysis. <https://doi.org/10.32614/CRAN.package.fda>
- Reynolds, A. P., Richards, G., De La Iglesia, B., & Rayward-Smith, V. J. (2006). Clustering Rules: A Comparison of Partitioning and Hierarchical Clustering Algorithms. *Journal of Mathematical Modelling and Algorithms*, 5(4), 475–504. <https://doi.org/10.1007/s10852-005-9022-1>
- Rietveld, A., & Gussenhoven, C. (1985). On the relation between pitch excursion size and prominence. *Journal of Phonetics*, 13(3), 299–308. [https://doi.org/10.1016/S0095-4470\(19\)30761-2](https://doi.org/10.1016/S0095-4470(19)30761-2)
- Ripley, B., & Venables, B. (2009). MASS: Support Functions and Datasets for Venables and Ripley’s MASS. <https://doi.org/10.32614/CRAN.package.MASS>
- Rissanen, J. (1978). Modeling by shortest data description. *Automatica*, 14(5), 465–471. [https://doi.org/10.1016/0005-1098\(78\)90005-5](https://doi.org/10.1016/0005-1098(78)90005-5)
- Rose, P. (1987). Considerations in the normalisation of the fundamental frequency of linguistic tone. *Speech Communication*, 6(4), 343–352. [https://doi.org/10.1016/0167-6393\(87\)90009-4](https://doi.org/10.1016/0167-6393(87)90009-4)
- Scheffers, M. T. M. (1983). Simulation of auditory analysis of pitch: An elaboration on the DWS pitch meter. *The Journal of the Acoustical Society of America*, 74(6), 1716–1725. <https://doi.org/10.1121/1.390280>
- Schubert, E., & Rousseeuw, P. J. (2019). Faster k-Medoids Clustering: Improving the PAM, CLARA, and CLARANS Algorithms. In G. Amato, C. Gennaro, V. Oria, & M. Radovanović (Eds.), *Similarity Search and Applications* (pp. 171–187, Vol. 11807). Springer International Publishing. https://doi.org/10.1007/978-3-030-32047-8_16

- Schubert, E., & Rousseeuw, P. J. (2021). Fast and eager k-medoids clustering: O (k) runtime improvement of the PAM, CLARA, and CLARANS algorithms. *Information Systems*, 101, 101804. <https://doi.org/10.1016/j.is.2021.101804>
- Scitovski, R., Sabo, K., Martínez-Álvarez, F., & Ungar, Š. (2021). *Cluster analysis and applications*. Springer Nature. <https://doi.org/10.1007/978-3-030-74552-3>
- Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3), 379–423. <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>
- Silverman, B. W. (1986). *Density estimation for statistics and data analysis*. Chapman & Hall/CRC.
- Trautmüller, H. (1990). Analytical expressions for the tonotopic sensory scale. *The Journal of the Acoustical Society of America*, 88(1), 97–100. <https://doi.org/10.1121/1.399849>
- Willems, L. F. (1987). Robust formant analysis for speech synthesis applications. *European Conference on Speech Technology*, 1250–1253. <https://doi.org/10.21437/ECST.1987-18>
- Winkelmann, R., Bombien, L., Scheffers, M., & Jochim, M. (2023). *Wrassp: Interface to the 'ASSP' library*. Manual.
- Xu, Y., & Sun, X. (2002). Maximum speed of pitch change and how it may relate to speech. *The Journal of the Acoustical Society of America*, 111(3), 1399–1413. <https://doi.org/10.1121/1.1445789>
- Zwicker, E. (1961). Subdivision of the Audible Frequency Range into Critical Bands (Frequenzgruppen). *The Journal of the Acoustical Society of America*, 33(2), 248–248. <https://doi.org/10.1121/1.1908630>

11. Index

- acceleration, 23
- acoustic measurements, 13
- alignment, 28
- application settings, 47
- ASCII, 12
- audio format, 6

- bending factor, 40
- between cluster variance, 41
- browser window, 8

- central tendency measure, 34, 35, 42
- cite Contour Clustering, 50
- cleaning, 21
- cluster analysis, 26
- clustering method, 28
- clustering variables, 26
- columns, 11, 20, 45
- contingency table, 7, 38
- convert audio, 47

- data (tab), 11
- data conventions, 44
- data protection, 50
- decimals, 46
- dendrogram, 29, 35, 36
- derivatives, 23
- df.l, 22
- display rows, 48

- distance metric, 27, 46
- duration, 16, 25
- dynamic time warping, 27, 28

- Euclidean distance, 27
- evaluation, 40, 46
- extrapolation, 14, 15, 20

- f0 ceiling, 13
- f0 correction, 23
- f0 fit, 13
- f0 floor, 13
- f0 scale, 23
- f0 smoothing, 14, 15
- f0 smoothing bandwidth, 14
- f0 time-step, 14
- file encoding, 12
- flagging, 38
- formants, 17, 25
- fPCA, 23, 24

- garbage collection, 49
- gender, 19
- Global Environment (R), 48

- HAC, 28
- hierarchical agglomerative clustering (HAC), 28

- IEEE float, 47

- information cost, 40
- intensity, 16, 24
- intensity peak, 19
- interpolation, 14, 15, 20
- interval tier, 6
- inversion risk, 30
- jerk, 23
- kernel density estimation (KDE), 14, 32
- lambda, 24
- license, 51, 53
- line aesthetics, 48
- linkage criterion, 29
- Lobanov-normalisation, 25
- logfile, 49
- long format, 11, 44, 45
- MASE, 27, 28
- maximum rate of change (f_0), 22
- MDL, 41
- mean standard error, 6, 38
- medoid, 31
- minimum description length (MDL), 40
- minimum interval duration, 14
- Modified Harmonic Sieve, 13
- Montreal Forced Aligner (MFA), 7
- multivariate clustering, 26
- number of clusters, 6, 35, 37–41
- number of measurement points, 14, 15, 45
- objects, 44, 48
- observations, 6
- octave-median rescaling, 23, 24
- optimizers (PAM), 31
- outliers, 15, 24, 27, 31, 35, 36, 39
- packages, 8, 27
- PAM, 31
- partitioning around medoids (PAM), 31
- Pearson correlation, 27, 28
- phonation, 13
- plot output, 37
- plot settings, 35
- principal component analysis, 23
- problems (cleaning), 21
- prototypicality, 37, 40, 42
- read folder, 12
- remove clusters, 39
- representations, 23, 26
- resolution, 33
- RMSE, 27
- run application, 8
- sampling, 20
- saved-folder, 48
- saving, 36, 48
- segmentation, 6
- singleton measure, 16, 23, 27, 32, 45
- SLA, 17
- smoothing bandwidth, 14, 22
- speakers, 21
- speed of processing, 12, 31, 46, 47, 49
- Split-Levinson algorithm (SLA), 17
- standardization, 23–25
- subsetting, 12, 13, 24, 29, 37–39
- summary table, 37
- table output, 37, 38
- textgrid, 5, 6, 12, 44
- time-series measure, 15, 16, 32, 45
- unit of analysis, 5, 6, 13, 15, 19, 40, 41

upload file (tab), 11, 44, 48
upload limit, 47
UTF-8, 12

velocity, 22, 23
vocal tract length, 19
vowel centroid, 25, 32, 46

Webmaus, 7
window size, 8
within cluster variance, 41
workflow, 8
www-folder, 20, 42

z-scoring, 23–25
zip file, 48